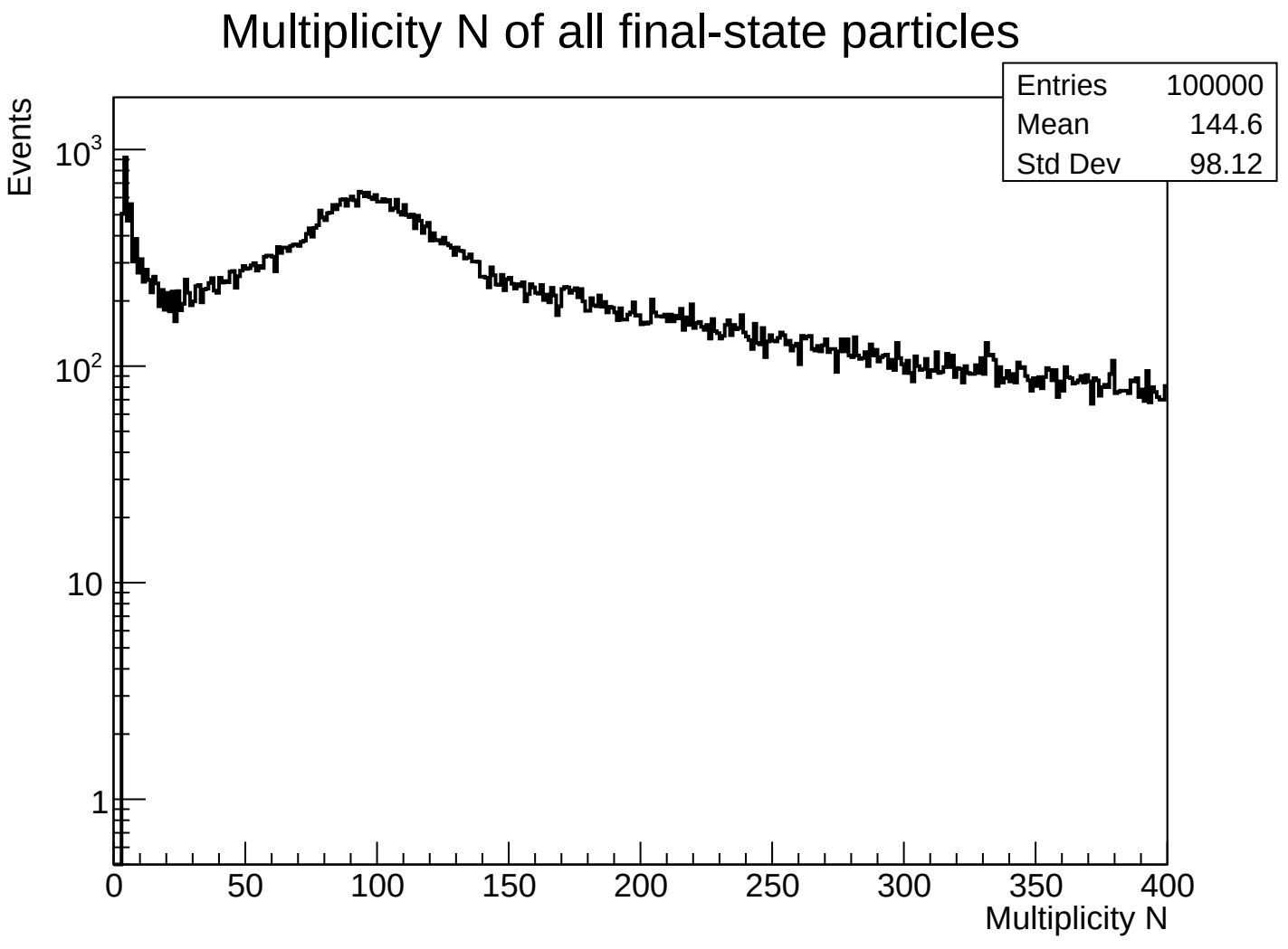
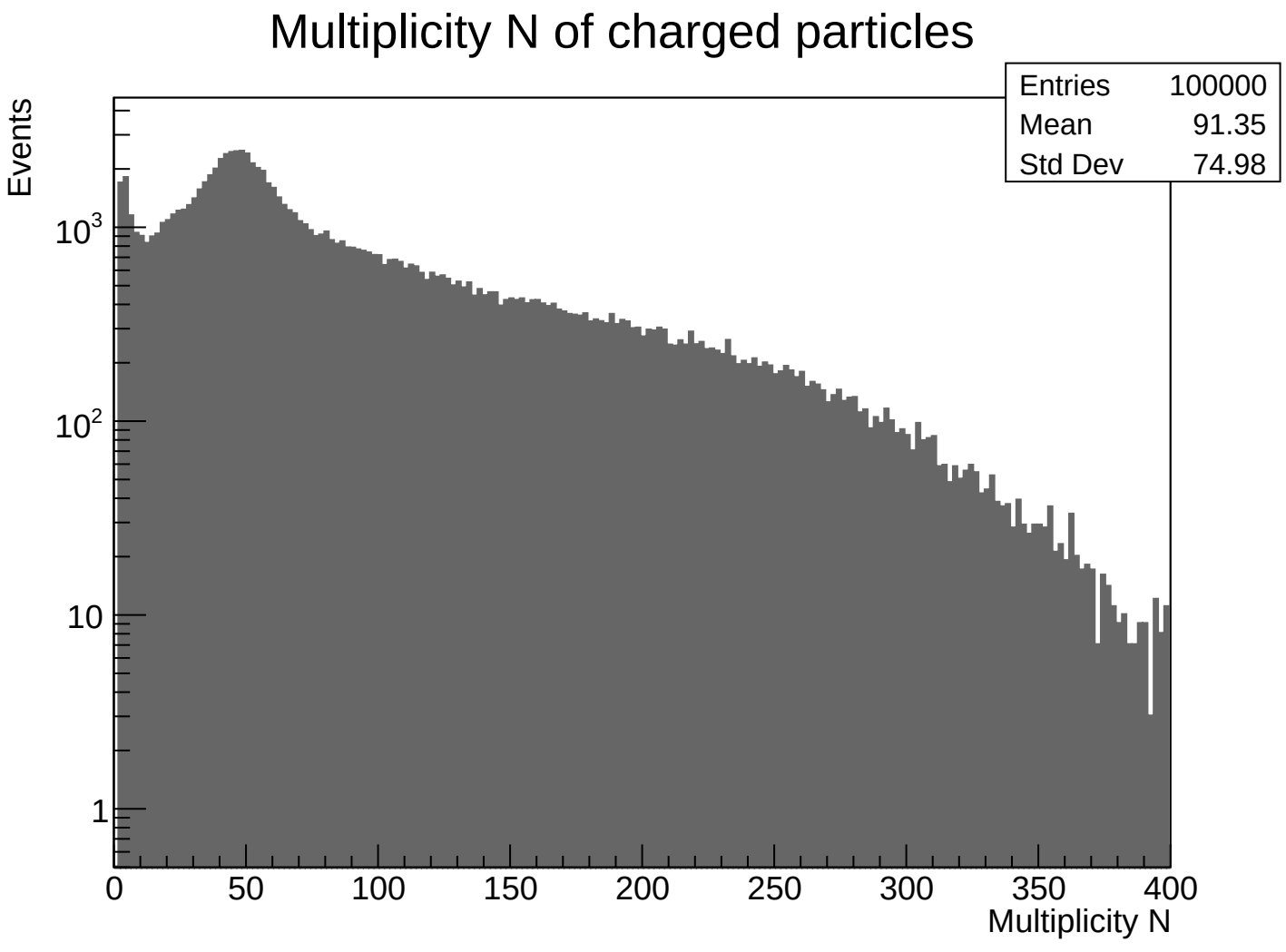
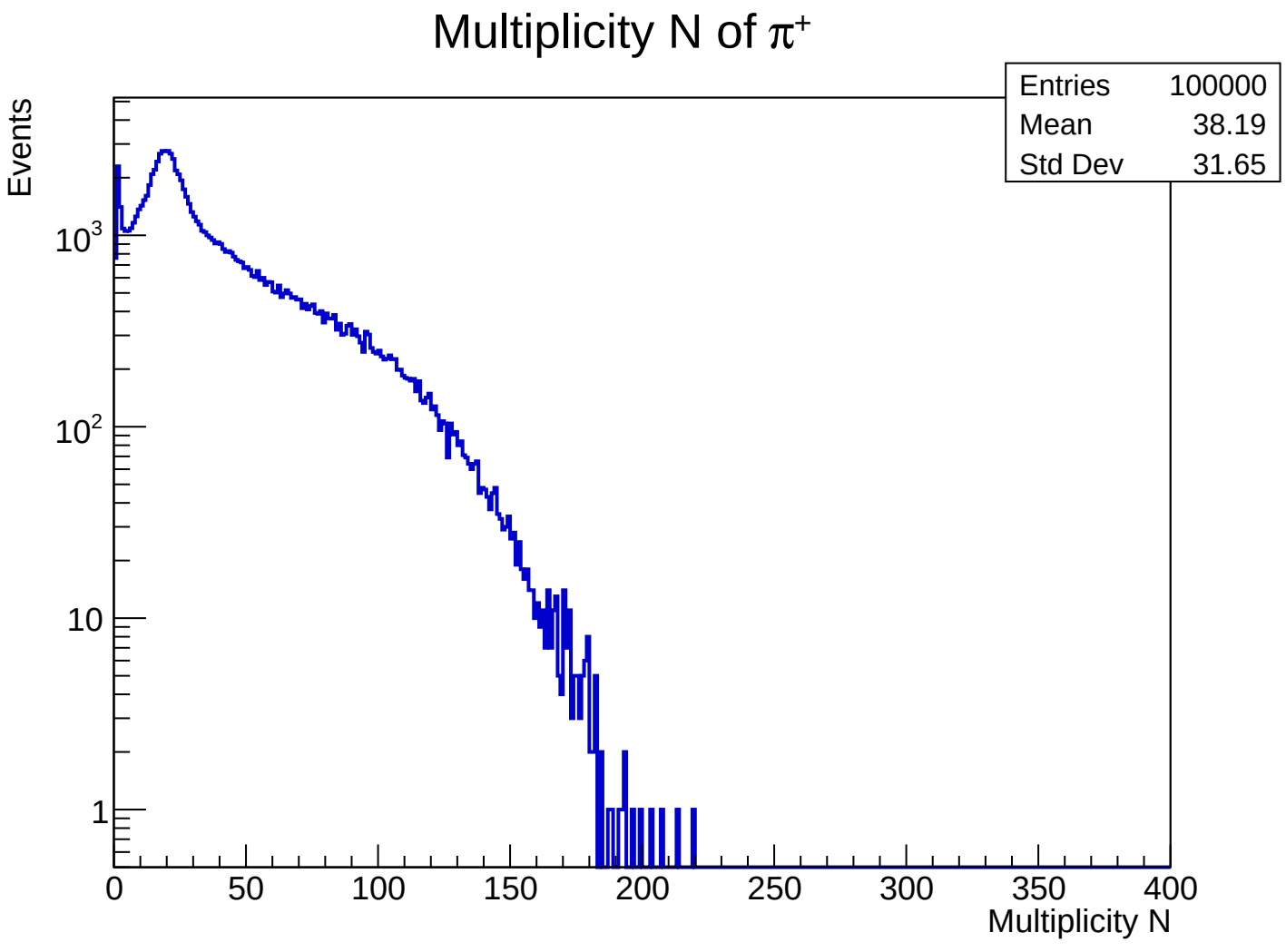
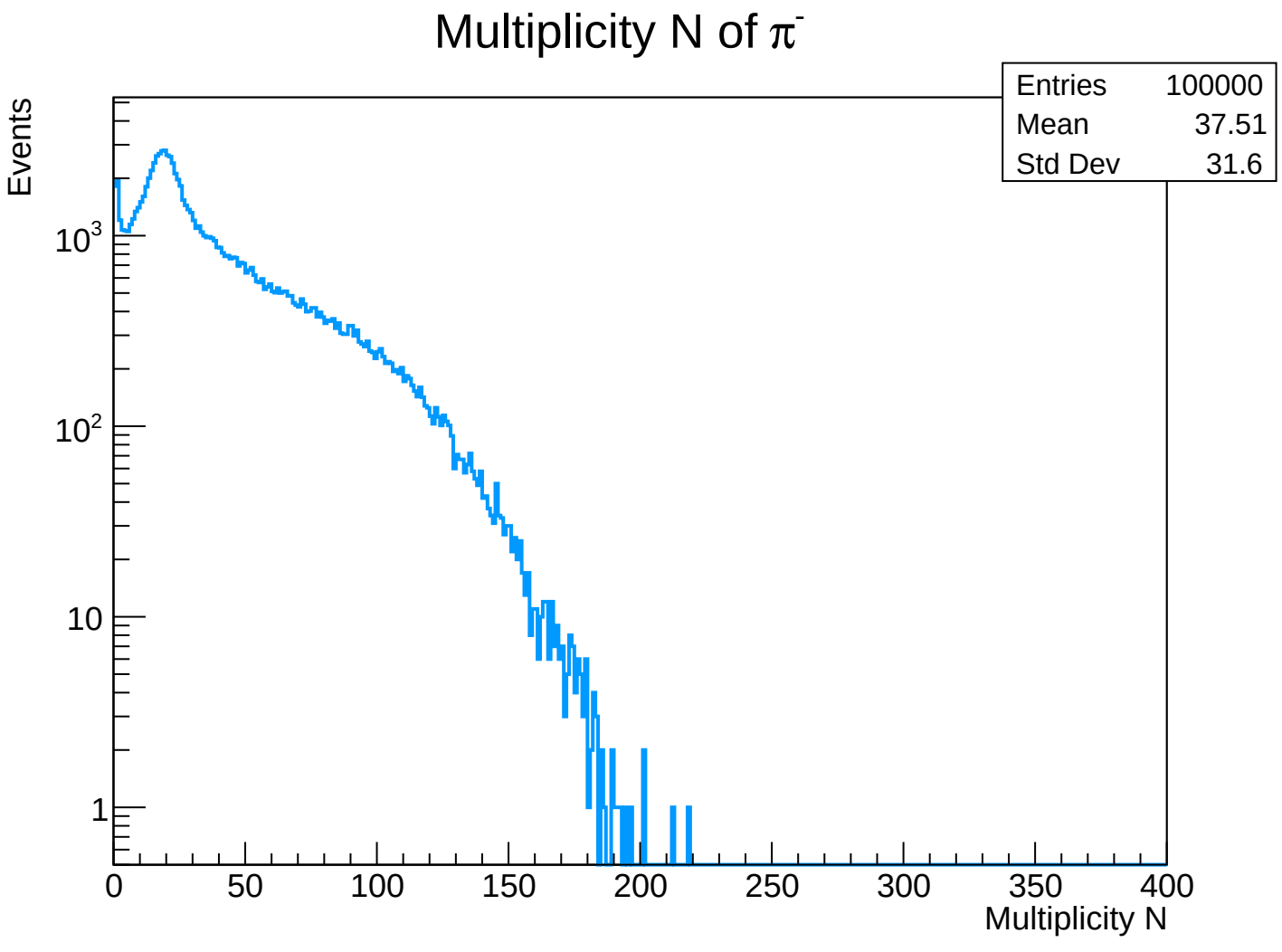
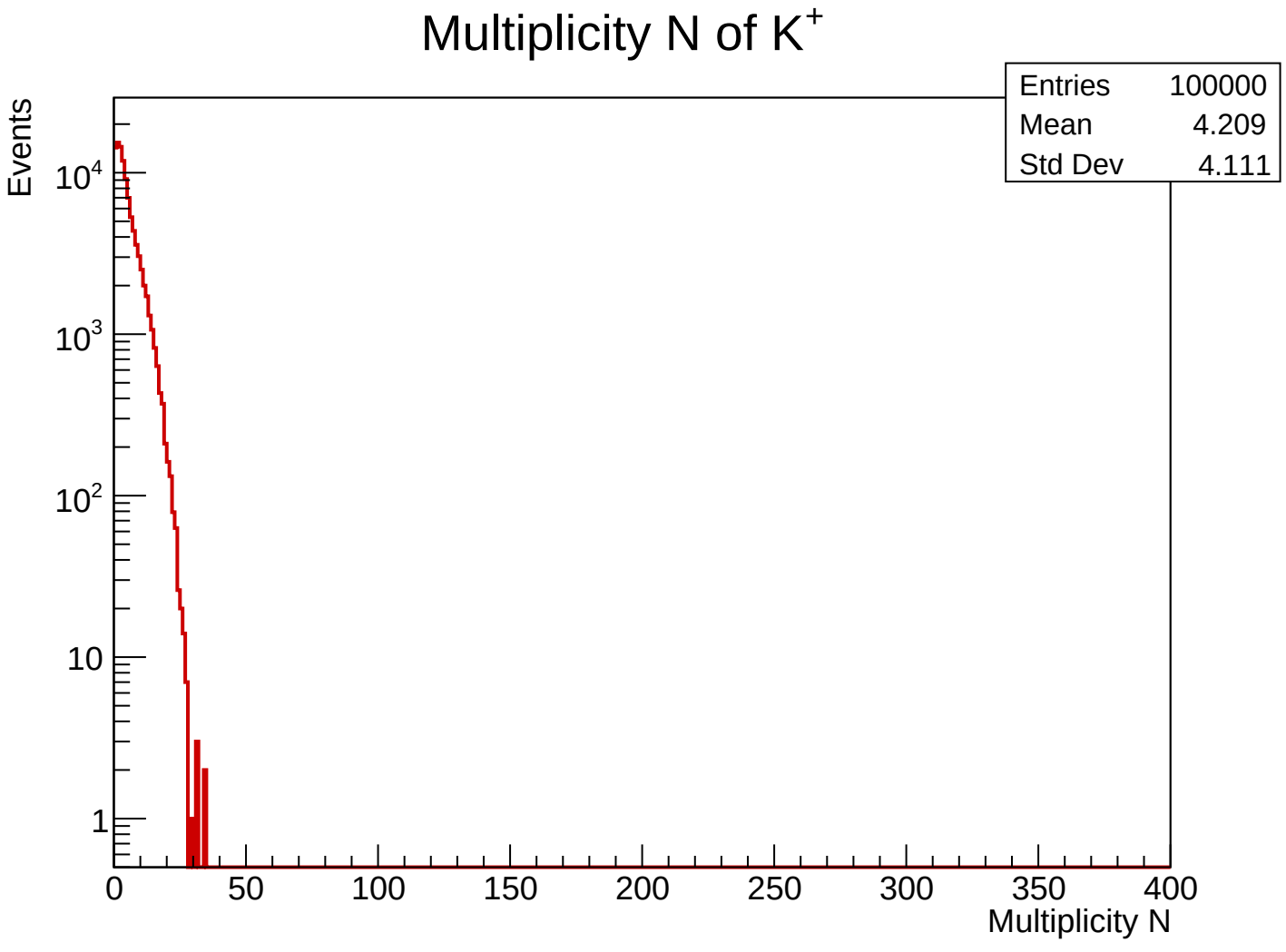
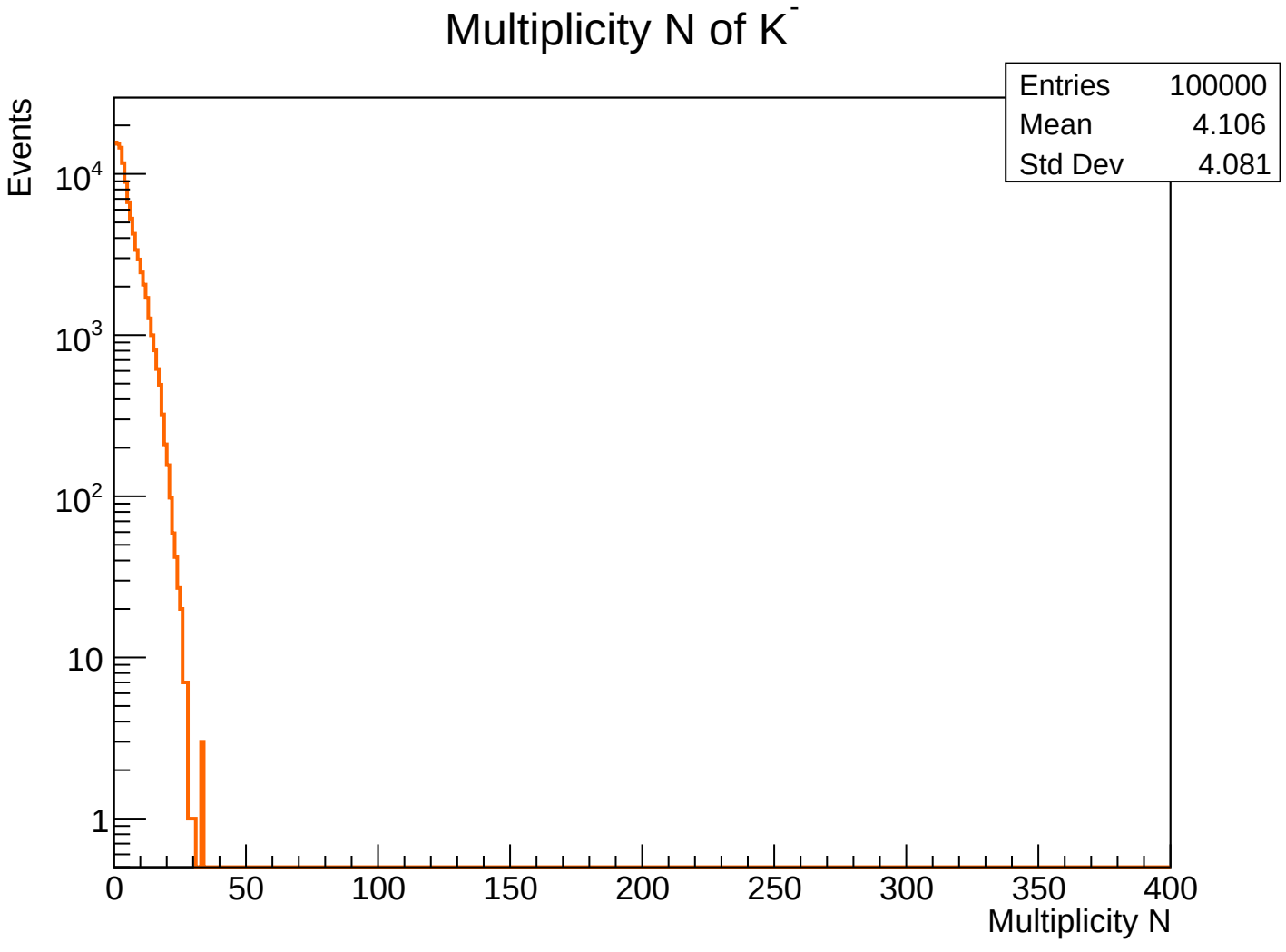

Multiplicity N of all final-state particles
Entries 100000
Mean 144.6
Std Dev 98.12
Events
10^3
10^2
10
1
0
50
100
150
200
250
300
350
400
Multiplicity N

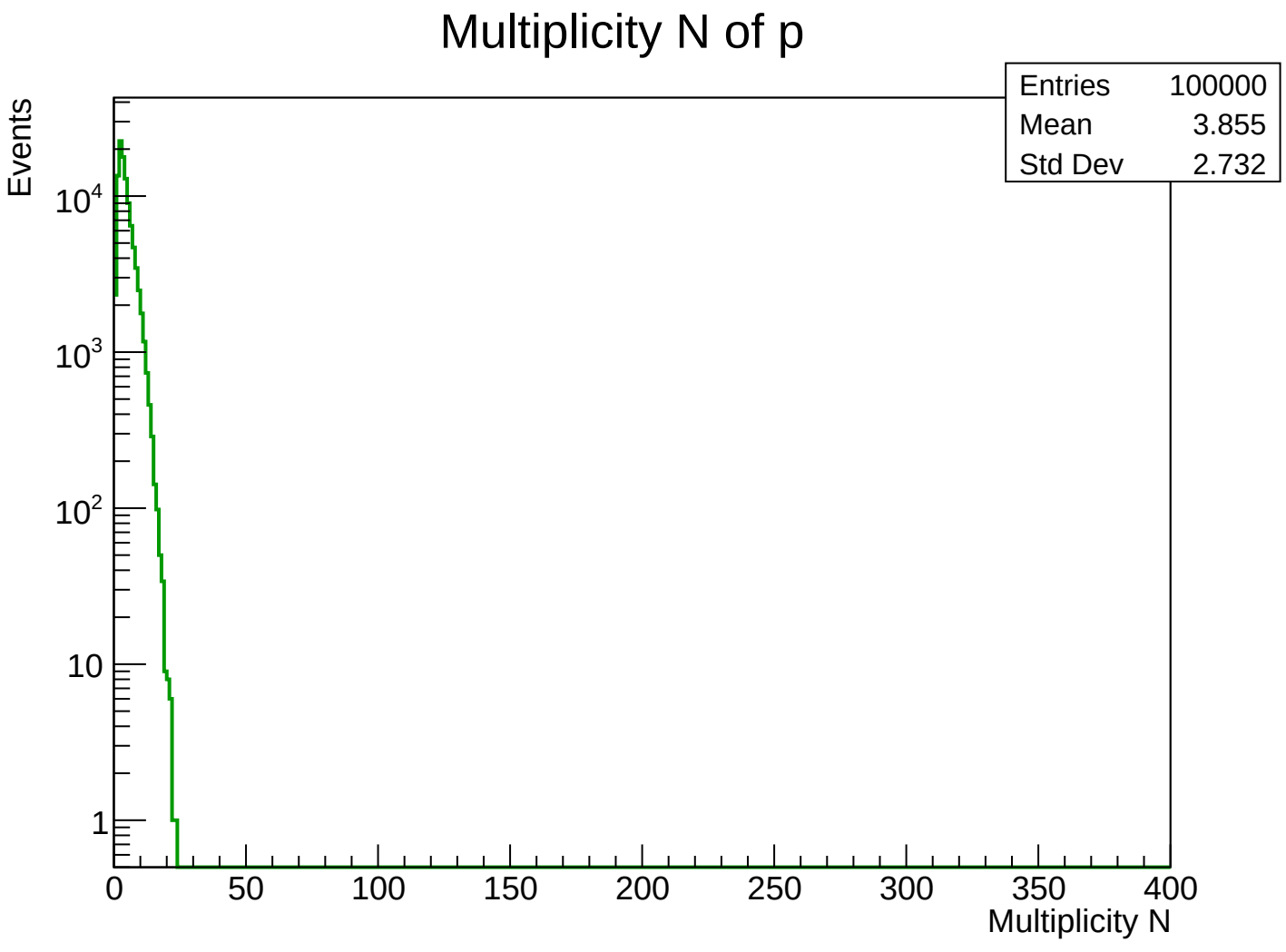

Multiplicity N of charged particles
Entries 100000
Mean 91.35
Std Dev 74.98
Events
10^3
10^2
10
1
0
50
100
150
200
250
300
350
400
Multiplicity N

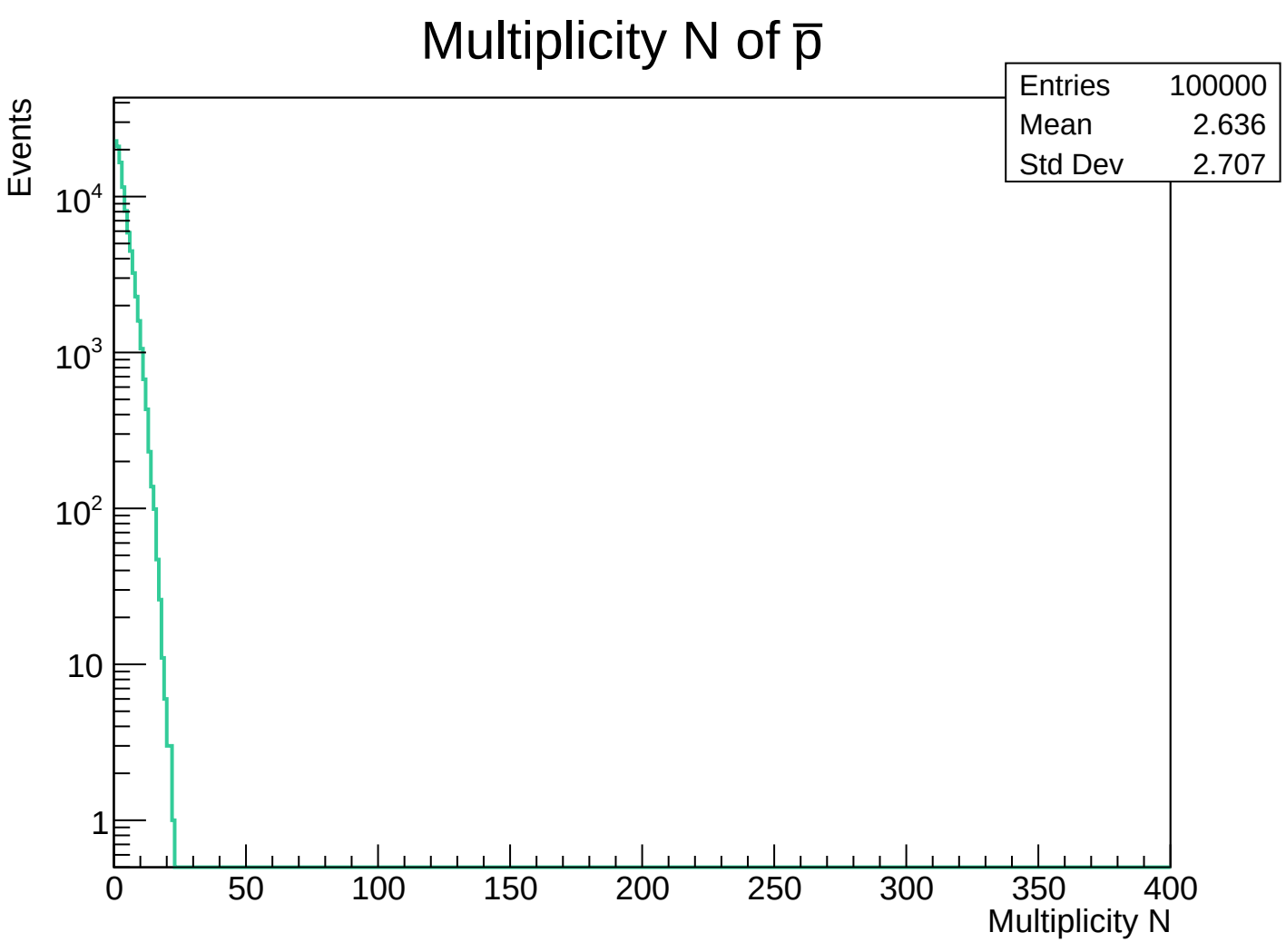

Multiplicity N of π^+
Entries 100000
Mean 38.19
Std Dev 31.65
Events
10^3
10^2
10
1
0
50
100
150
200
250
300
350
400
Multiplicity N

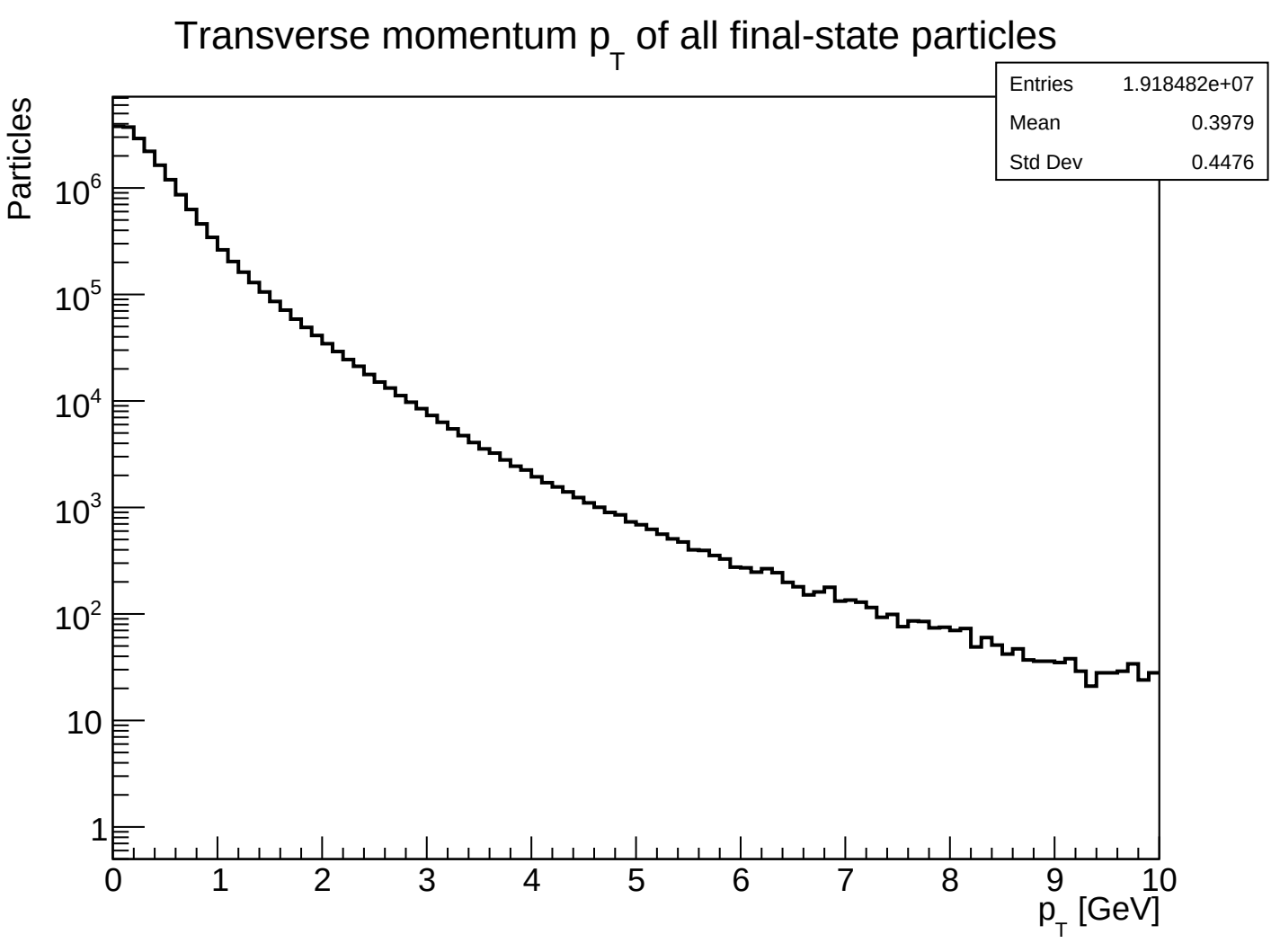

Multiplicity N of π^-
Entries 100000
Mean 37.51
Std Dev 31.6
Events
10^3
10^2
10
1
0
50
100
150
200
250
300
350
400
Multiplicity N

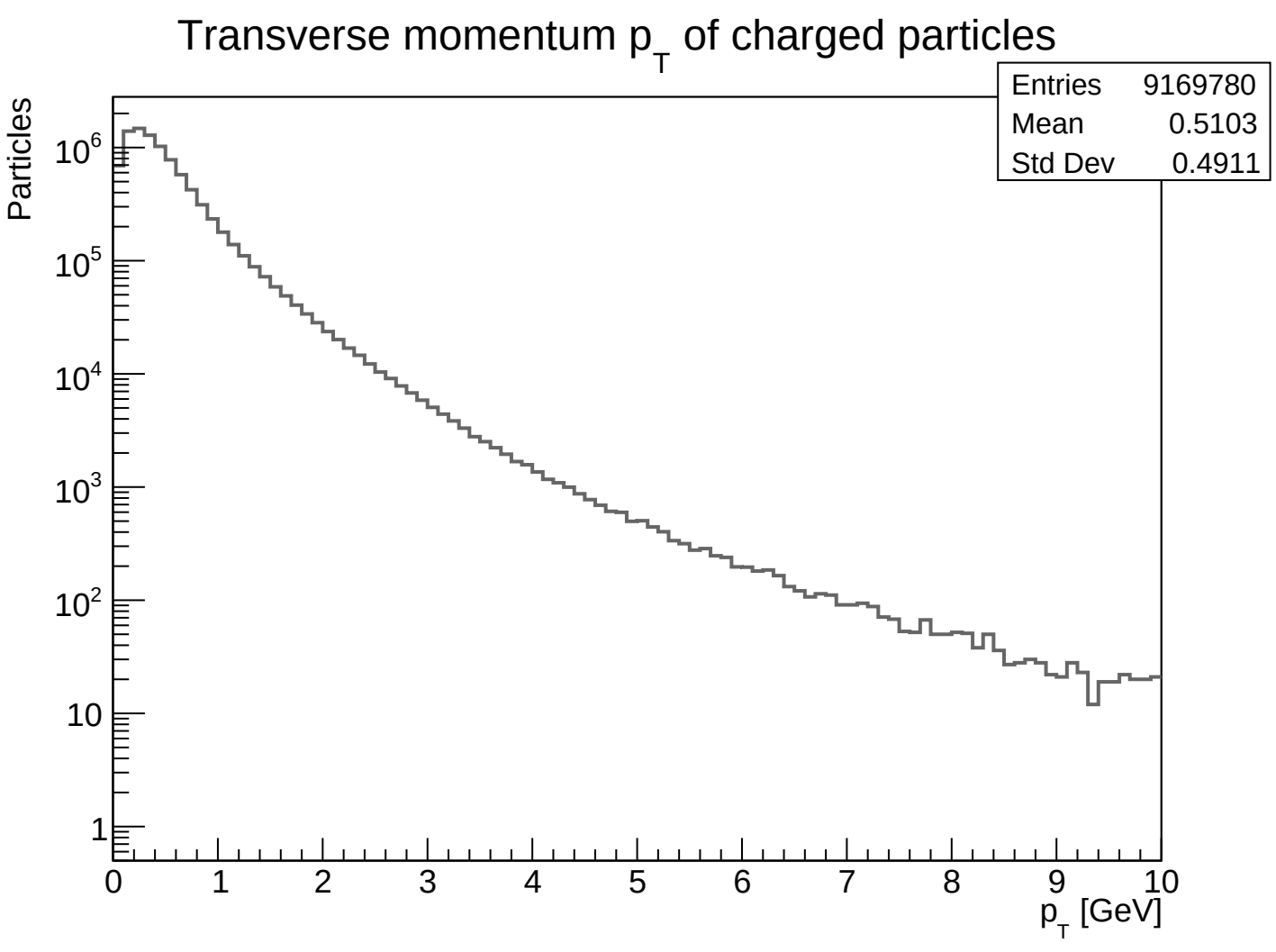

Multiplicity N of K+
Entries 100000
Mean 4.209
Std Dev 4.111
Events
10^4
10^3
10^2
10
1
0
50
100
150
200
250
300
350
400
Multiplicity N

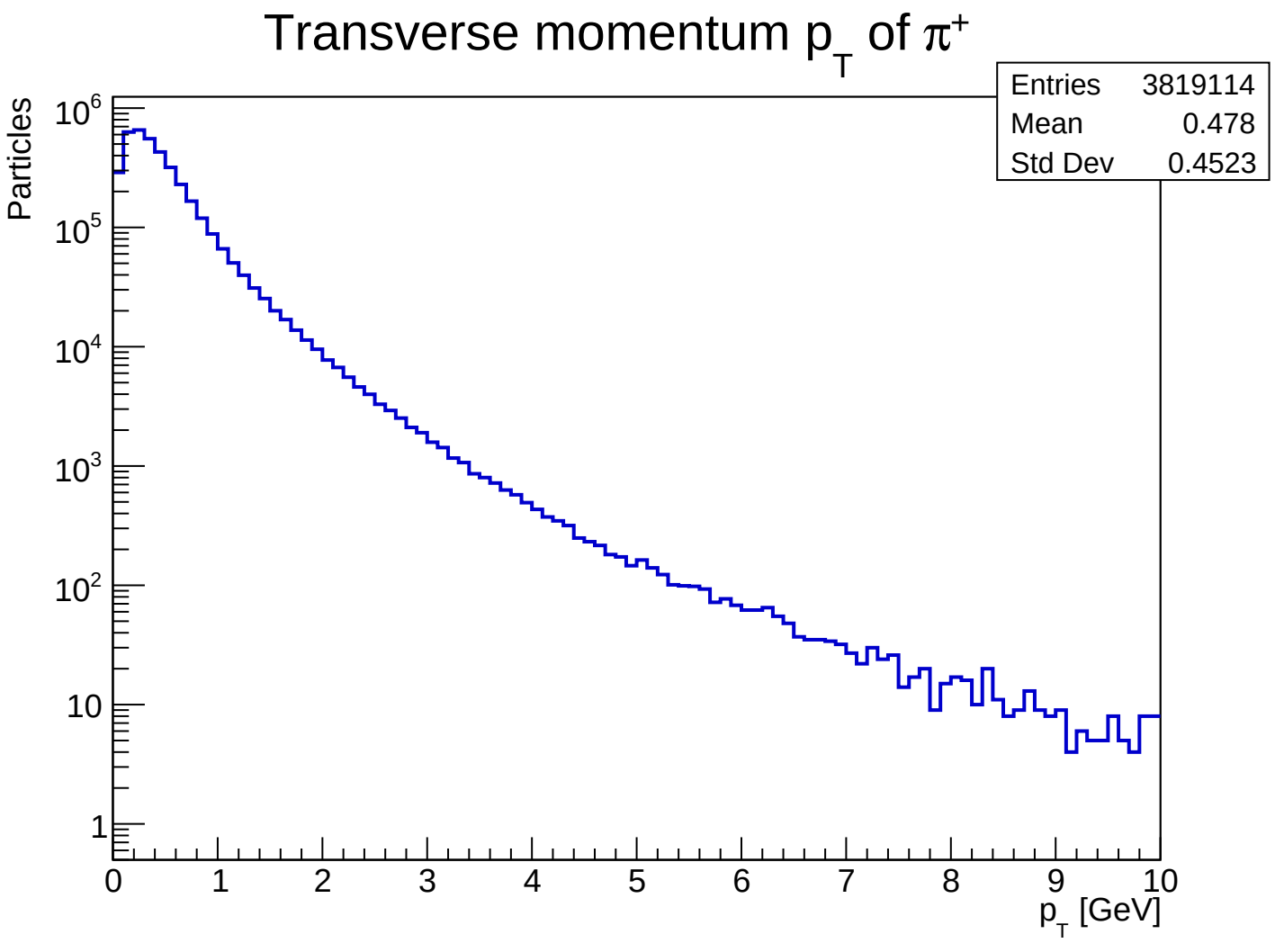

Multiplicity N of K-
Entries 100000
Mean 4.106
Std Dev 4.081
Events
10^4
10^3
10^2
10
1
0
50
100
150
200
250
300
350
400
Multiplicity N

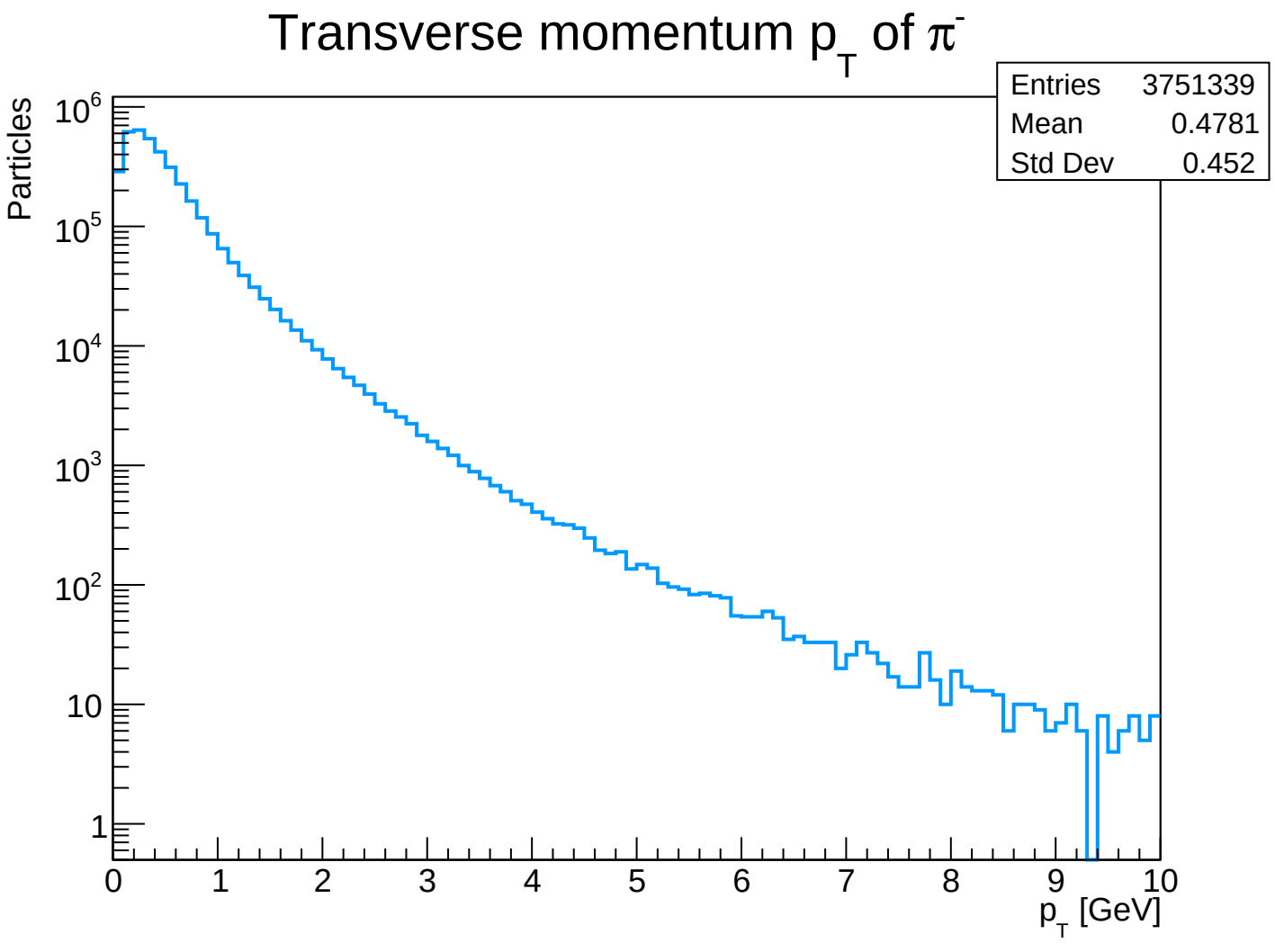

Multiplicity N of p
Entries 100000
Mean 3.855
Std Dev 2.732
Events
10^4
10^3
10^2
10
1
0
50
100
150
200
250
300
350
400
Multiplicity N

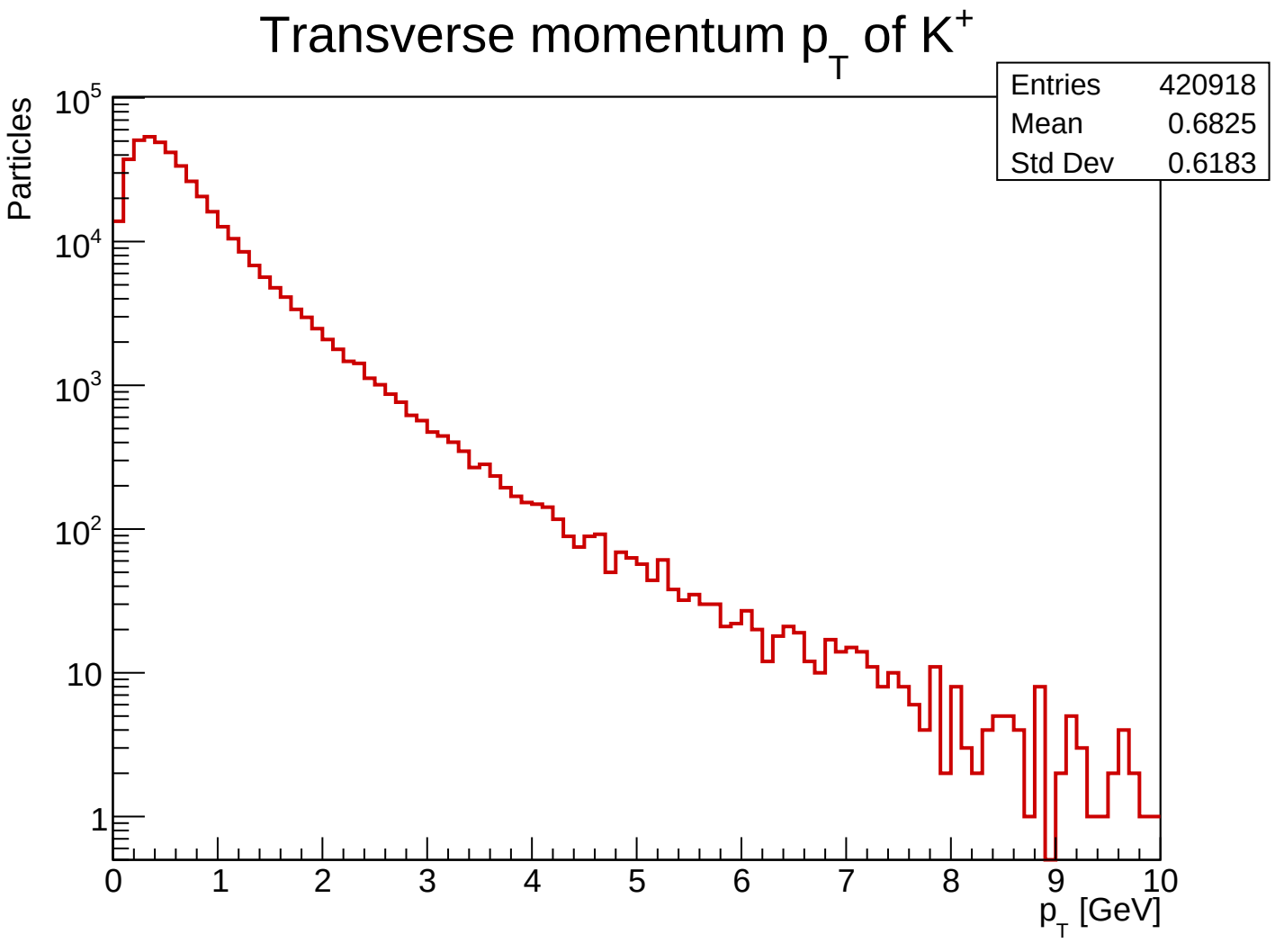

Multiplicity N of $\bar{p}$
Entries 100000
Mean 2.636
Std Dev 2.707
Events
10^4
10^3
10^2
10
1
0
50
100
150
200
250
300
350
400
Multiplicity N

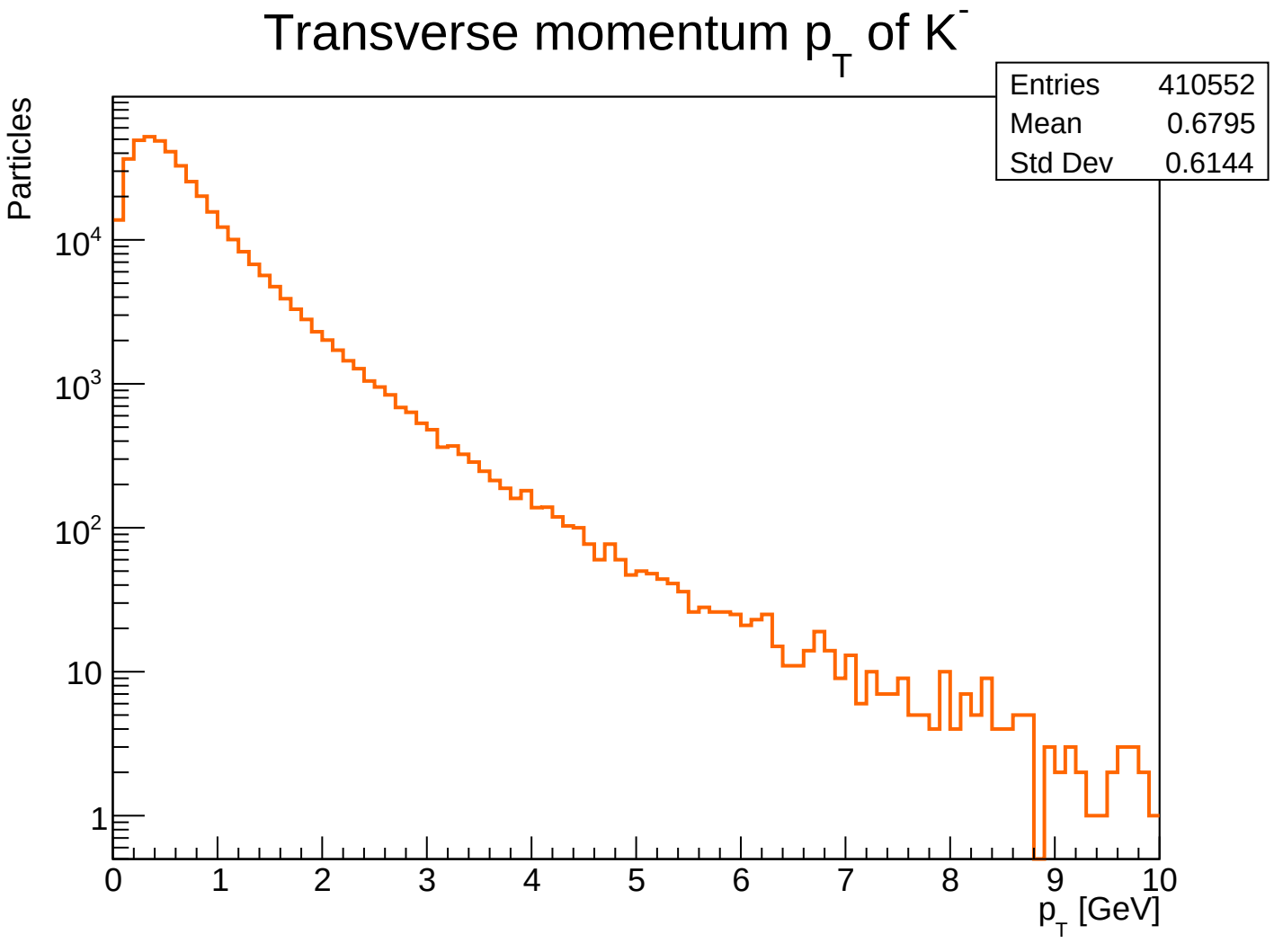

Transverse momentum p_T of all final-state particles
Entries 1.918482e+07
Mean 0.3979
Std Dev 0.4476
Particles
10^6
10^5
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8
9
10
p_T [GeV]

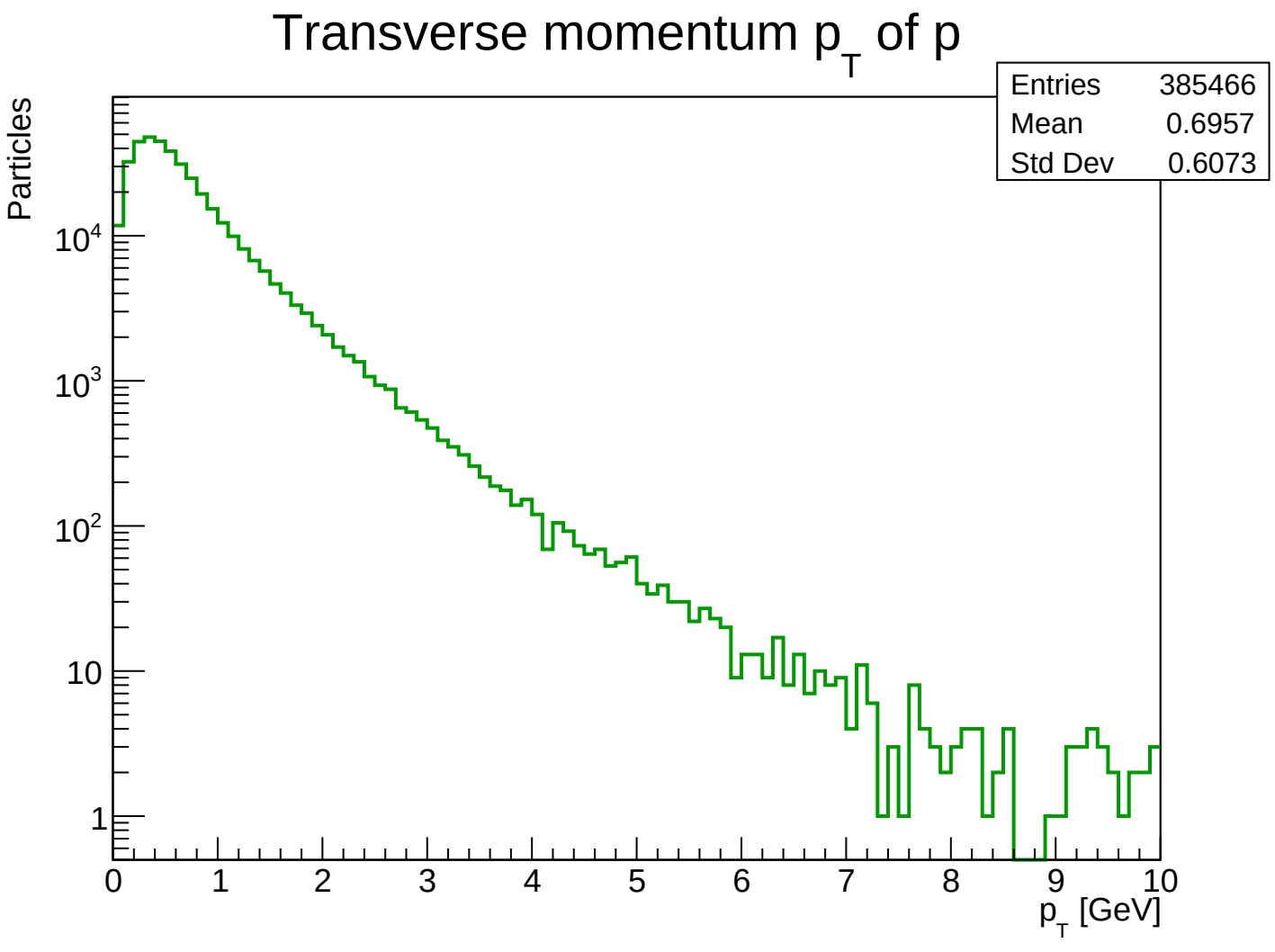

Transverse momentum p_T of charged particles
Entries 9169780
Mean 0.5103
Std Dev 0.4911
Particles
10^6
10^5
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8
9
10
p_T [GeV]

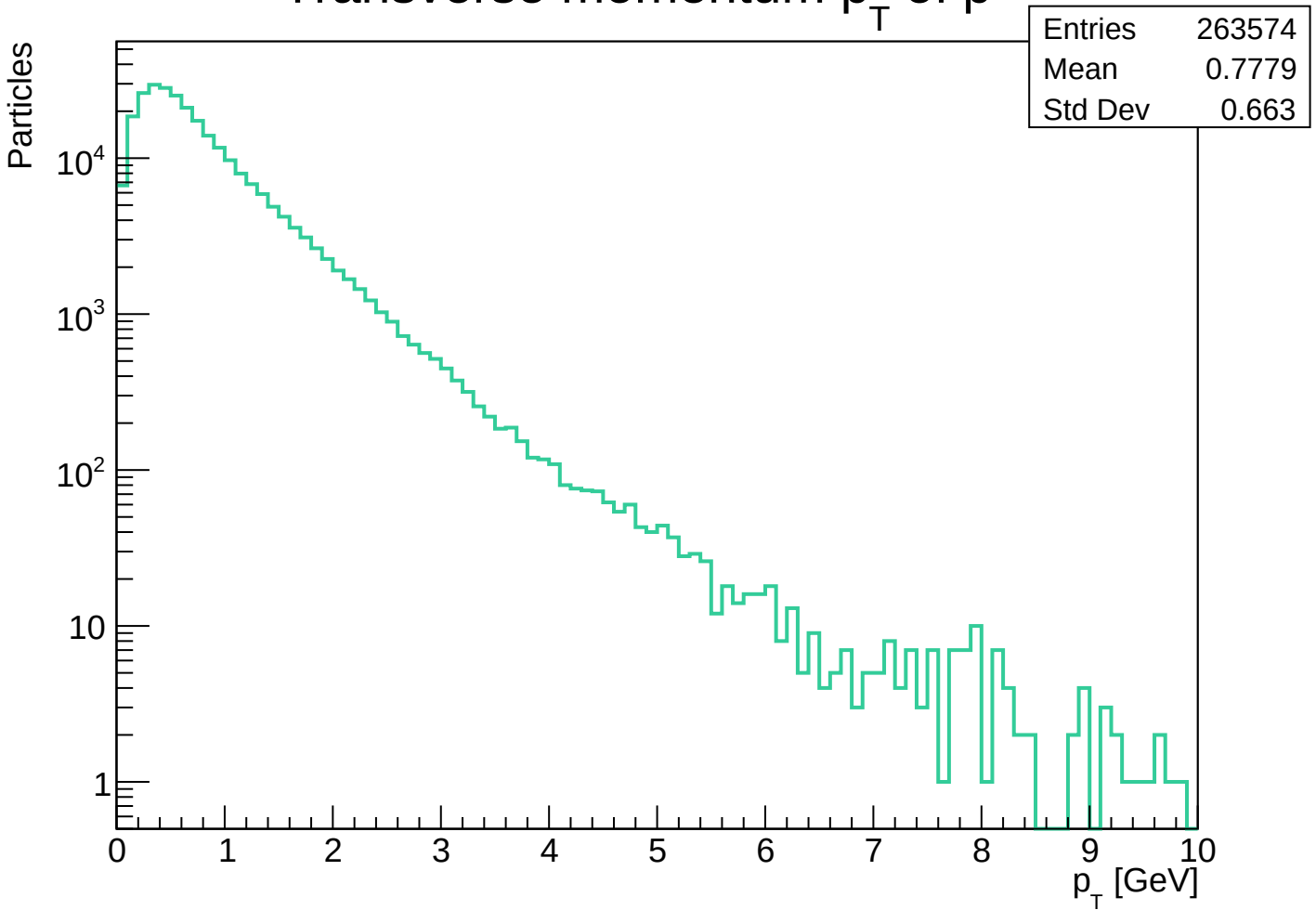

Transverse momentum p_T of π^+
Entries 3819114
Mean 0.478
Std Dev 0.4523
Particles
10^6
10^5
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8
9
10
p_T [GeV]

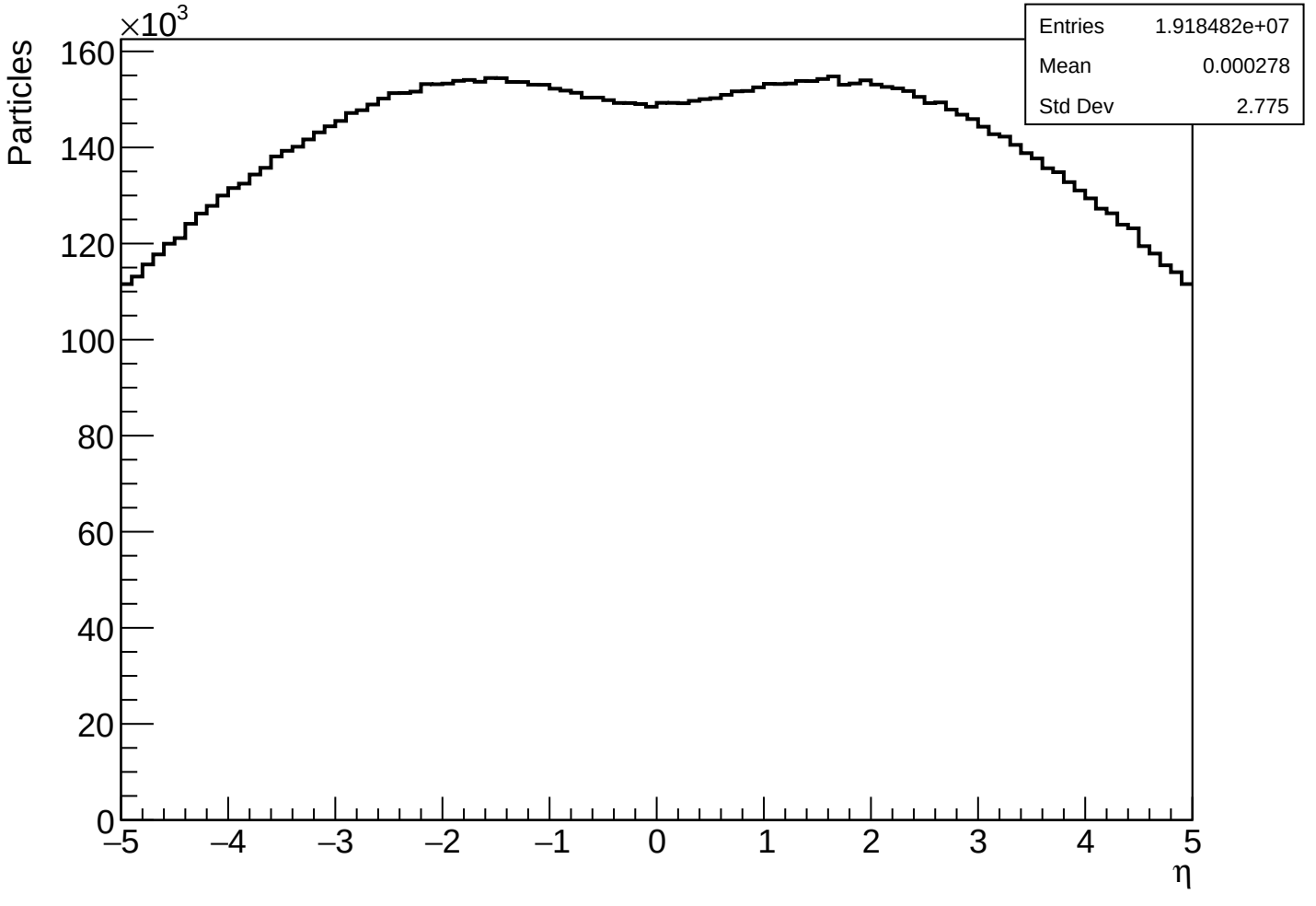

Transverse momentum p_T of π^-
Entries 3751339
Mean 0.4781
Std Dev 0.452
Particles
10^6
10^5
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8
9
10
p_T [GeV]

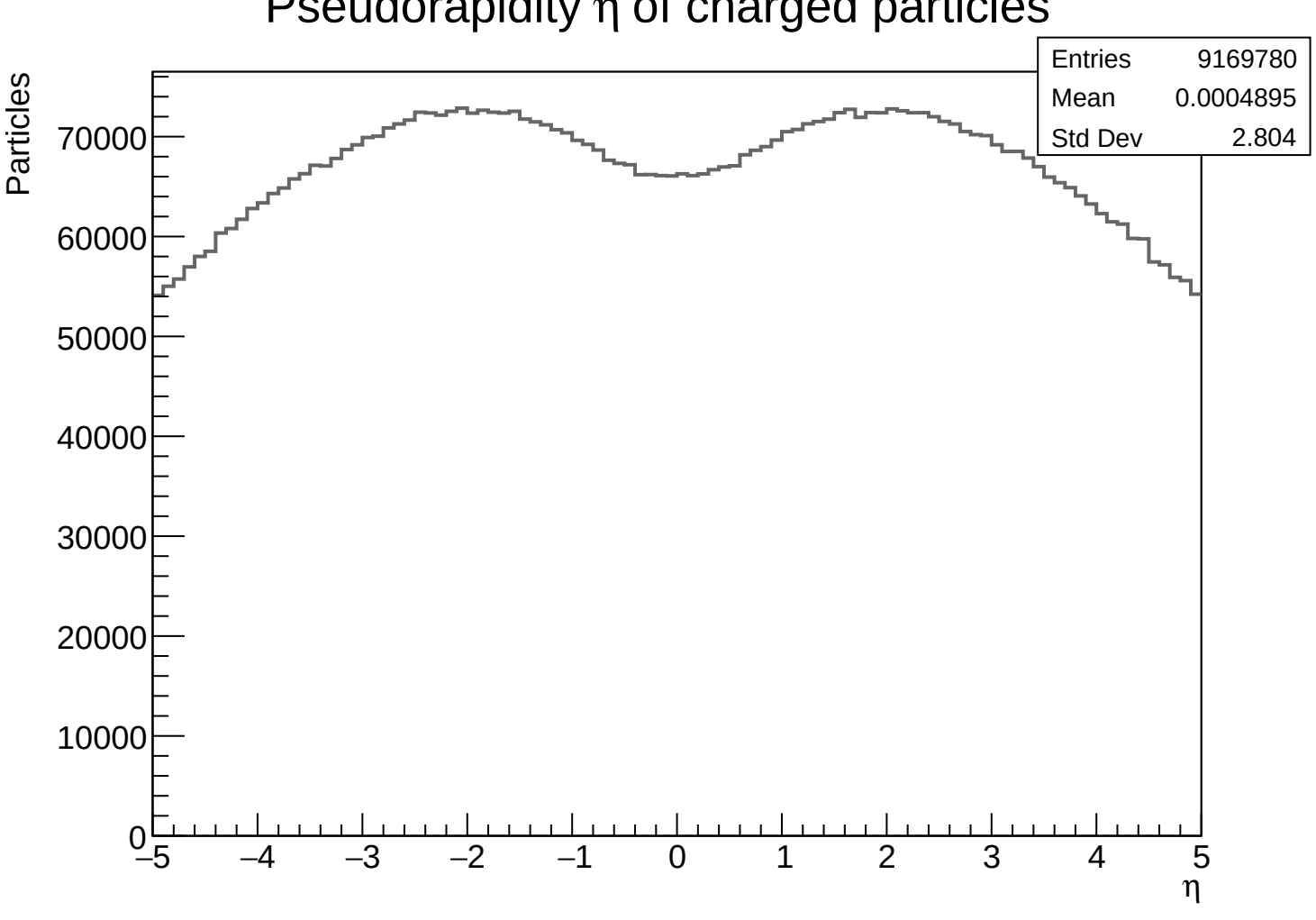

Transverse momentum p_T of K^+
Entries 420918
Mean 0.6825
Std Dev 0.6183
Particles
10^5
10^4
10^3
10^2
10
1
0 1 2 3 4 5 6 7 8 9 10
p_T [GeV]

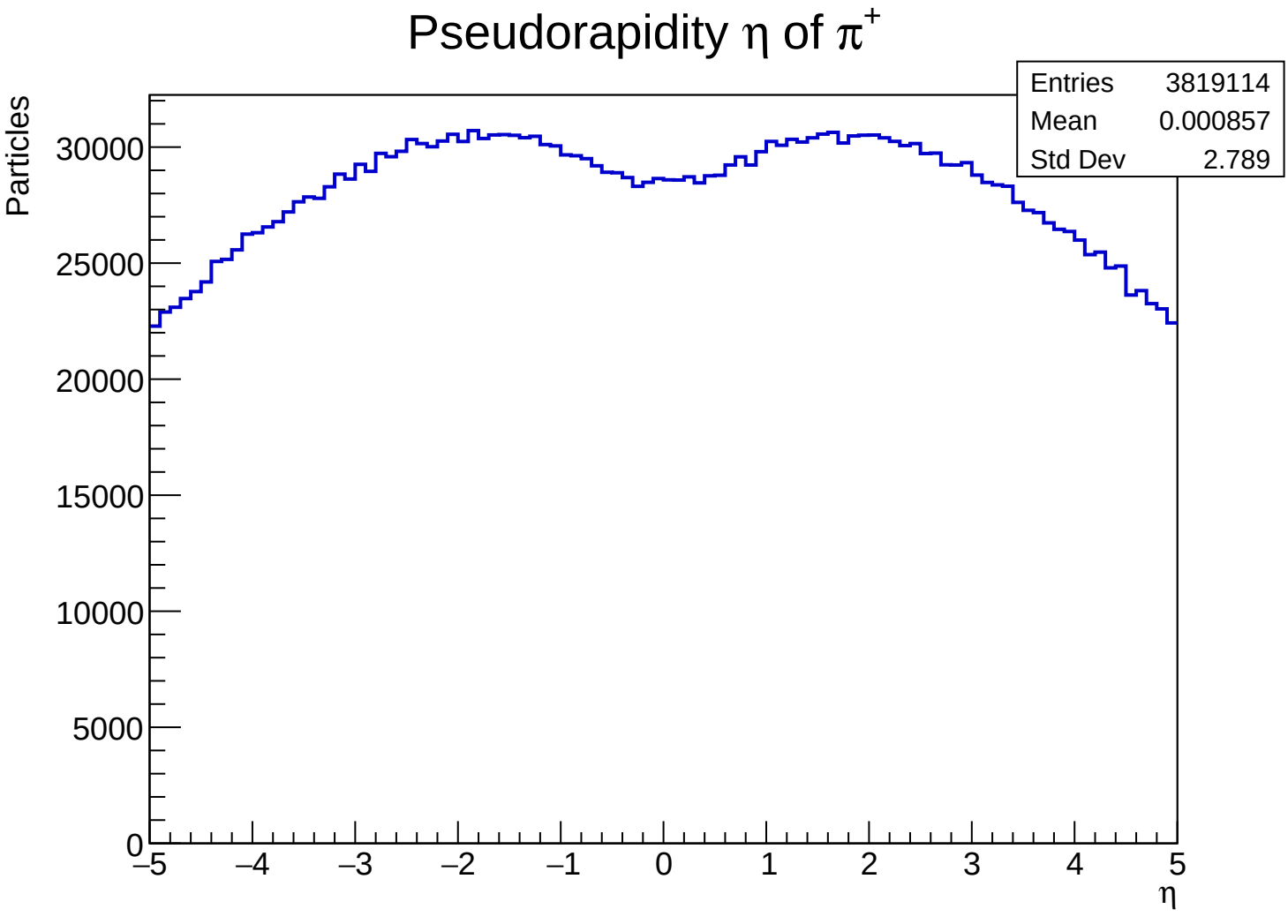

Transverse momentum p_T of K^-
Entries 410552
Mean 0.6795
Std Dev 0.6144
Particles
10^4
10^3
10^2
10
1
0 1 2 3 4 5 6 7 8 9 10
p_T [GeV]

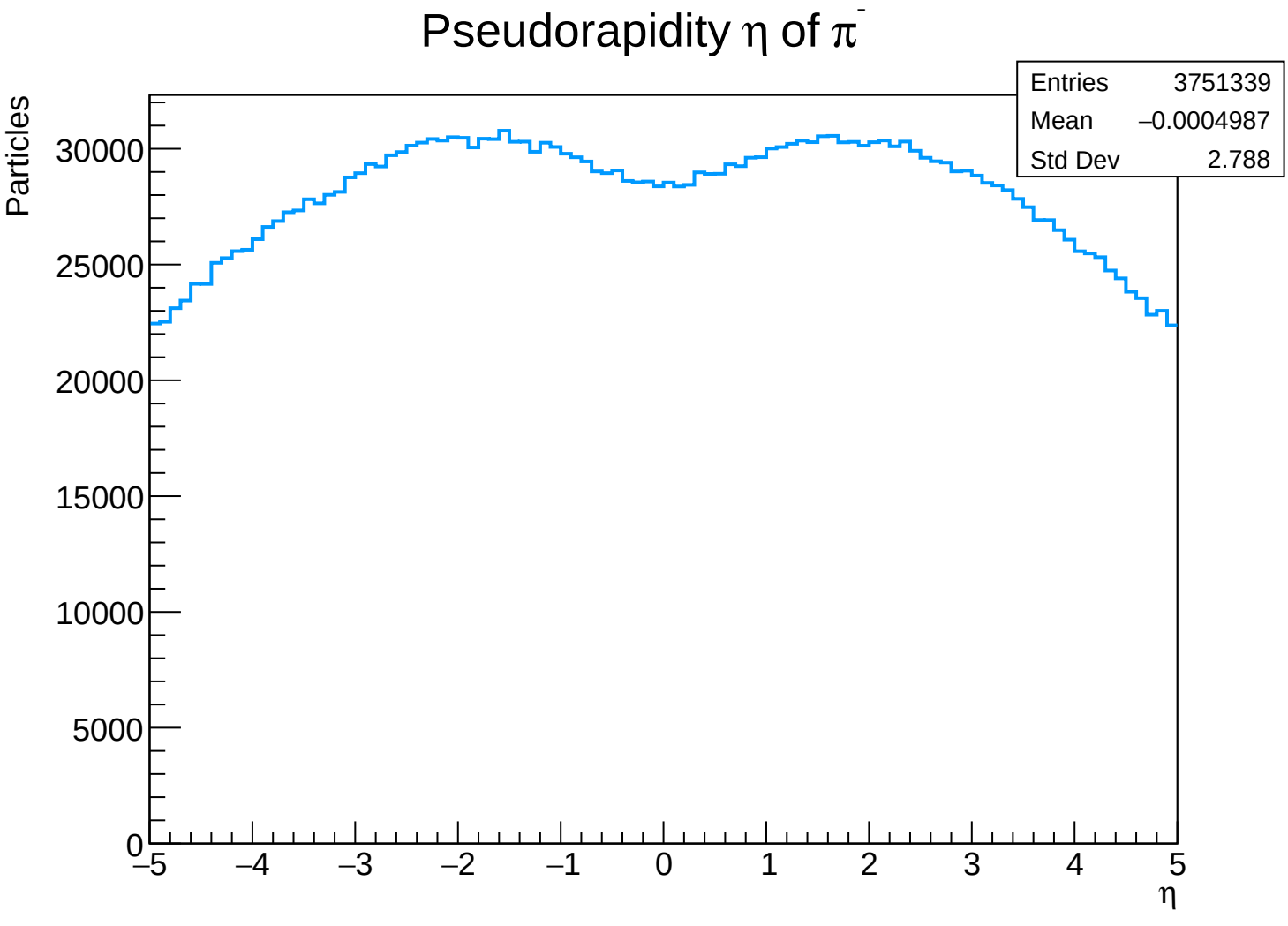

Transverse momentum p_T of p
Entries 385466
Mean 0.6957
Std Dev 0.6073
Particles
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8
9
10
p_T [GeV]

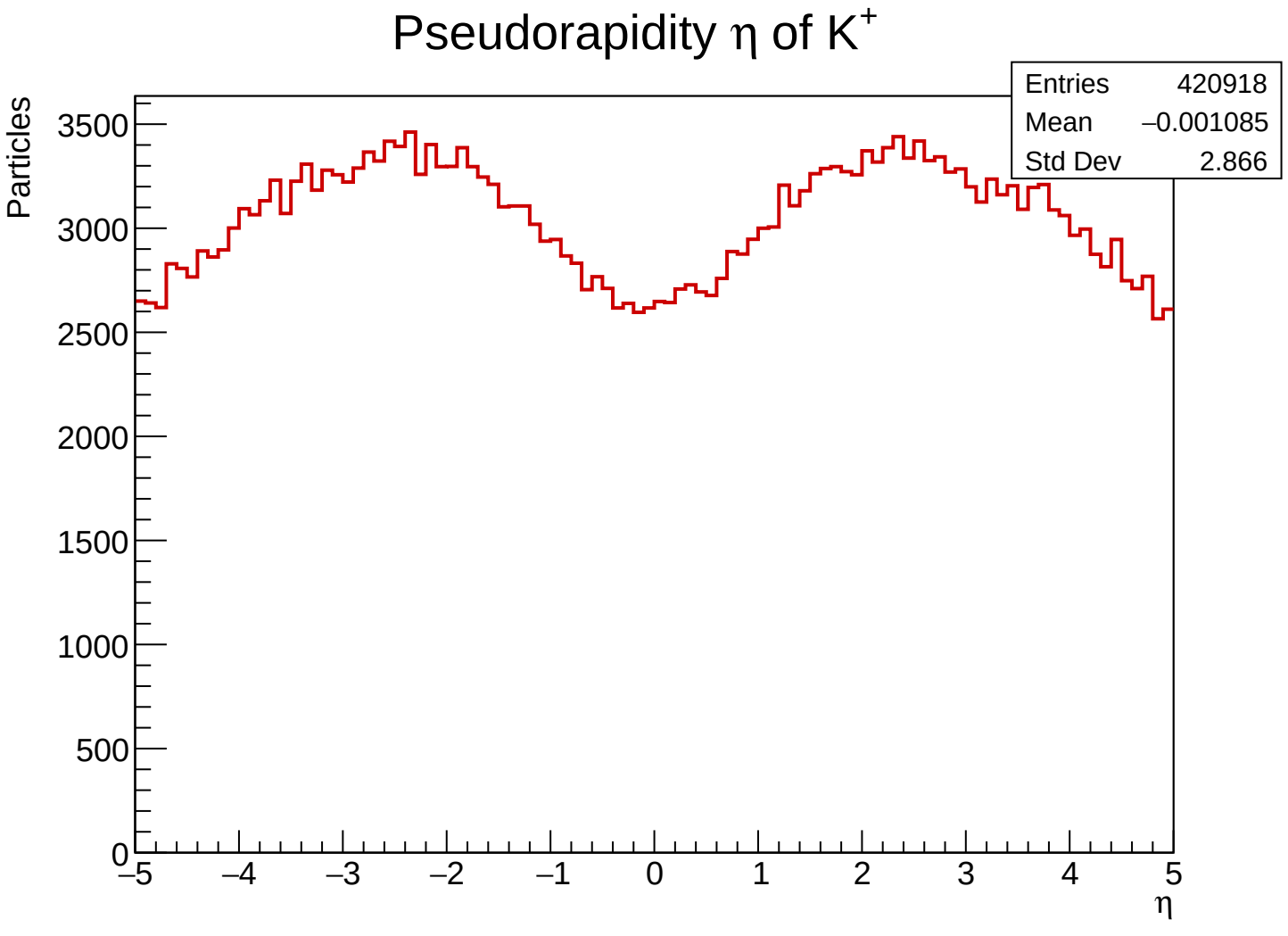

Transverse momentum p_T of $\bar{p}$
Entries 263574
Mean 0.7779
Std Dev 0.663
Particles
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8
9
10
p_T [GeV]

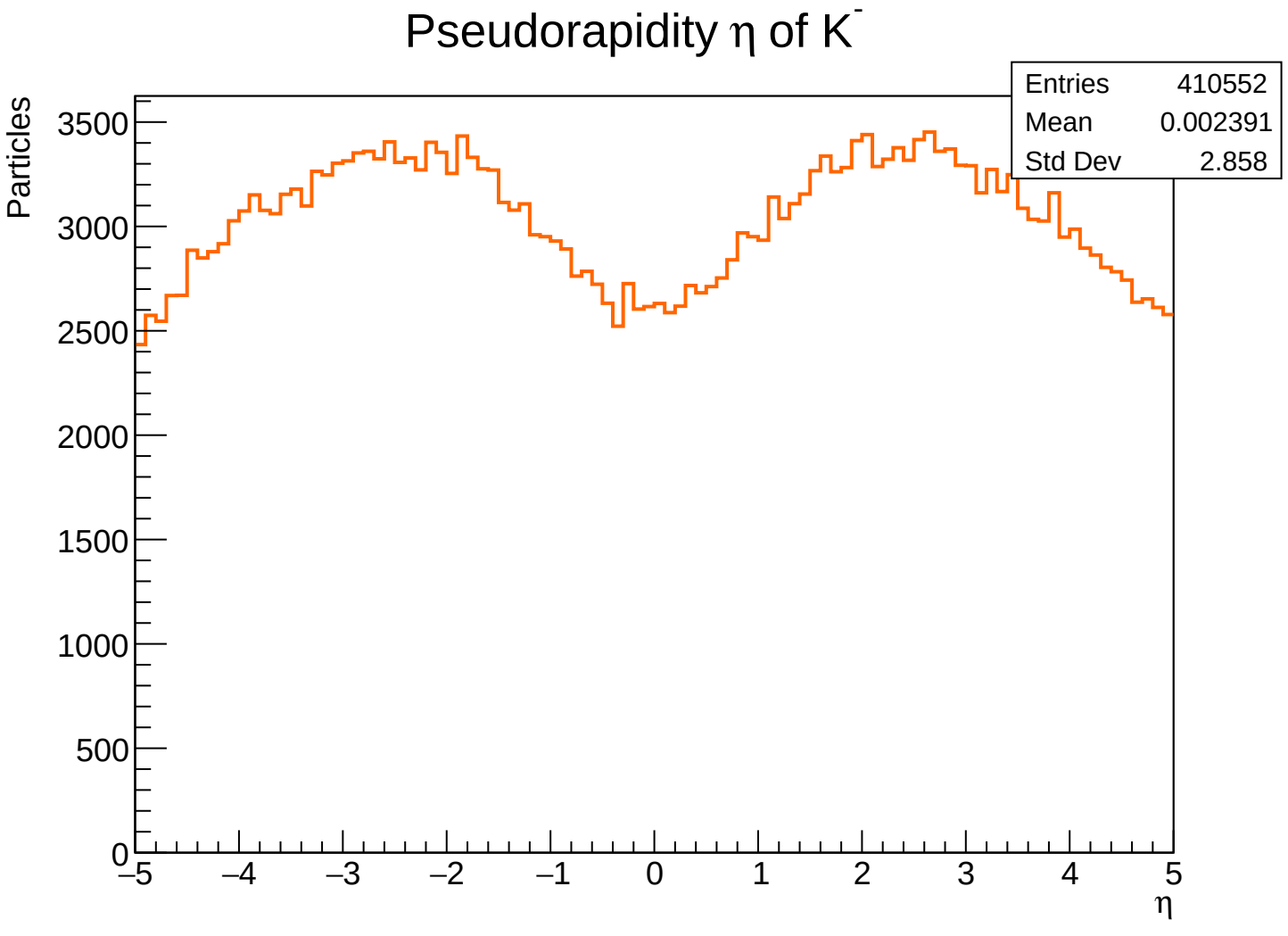

Pseudorapidity η of all final-state particles
Entries 1.918482e+07
Mean 0.000278
Std Dev 2.775
×10^3
Particles
160
140
120
100
80
60
40
20
0
–5
–4
–3
–2
–1
0
1
2
3
4
5
η

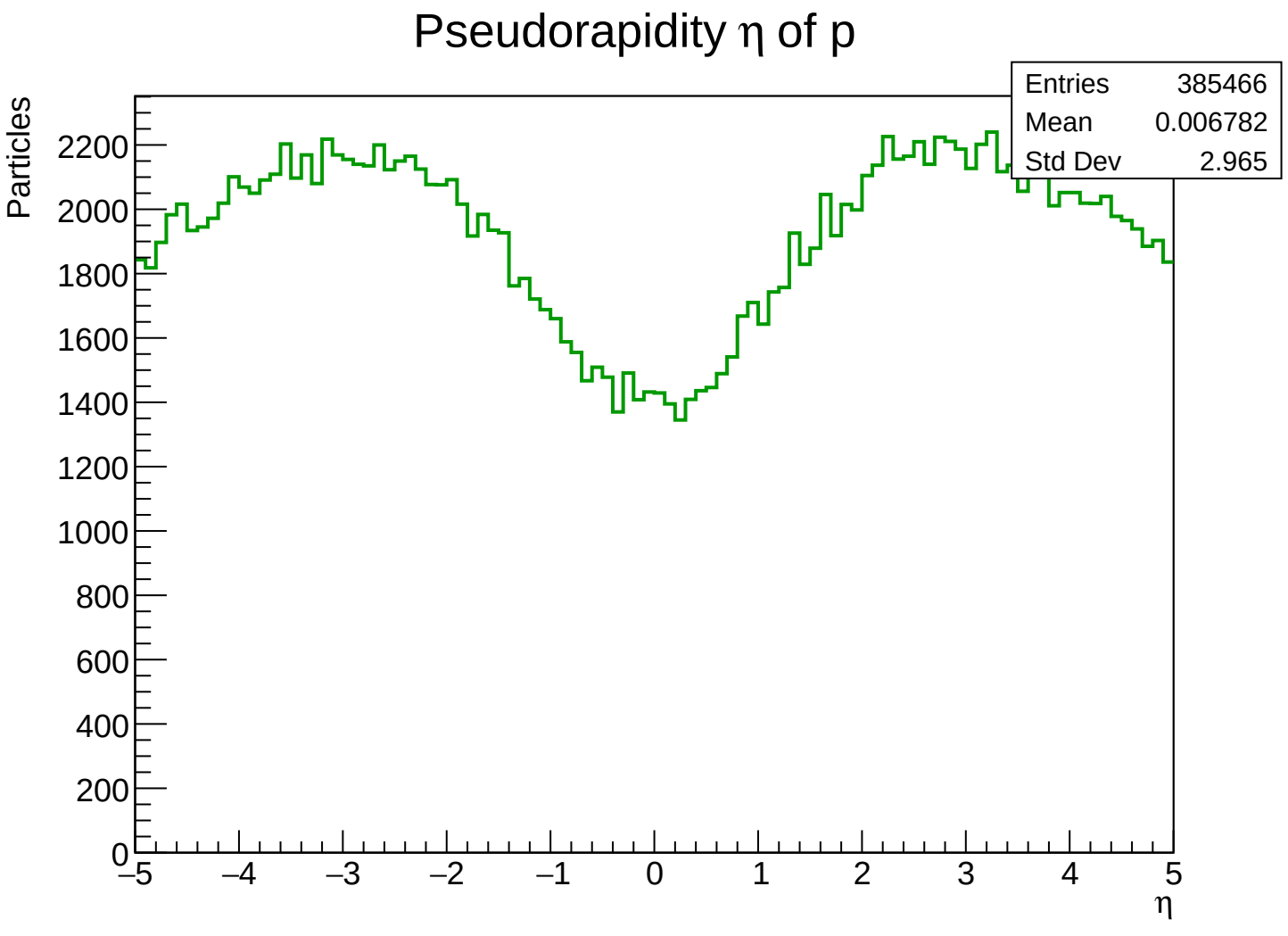

Pseudorapidity η of charged particles
Entries 9169780
Mean 0.0004895
Std Dev 2.804
Particles
70000
60000
50000
40000
30000
20000
10000
0
–5
–4
–3
–2
–1
0
1
2
3
4
5
η

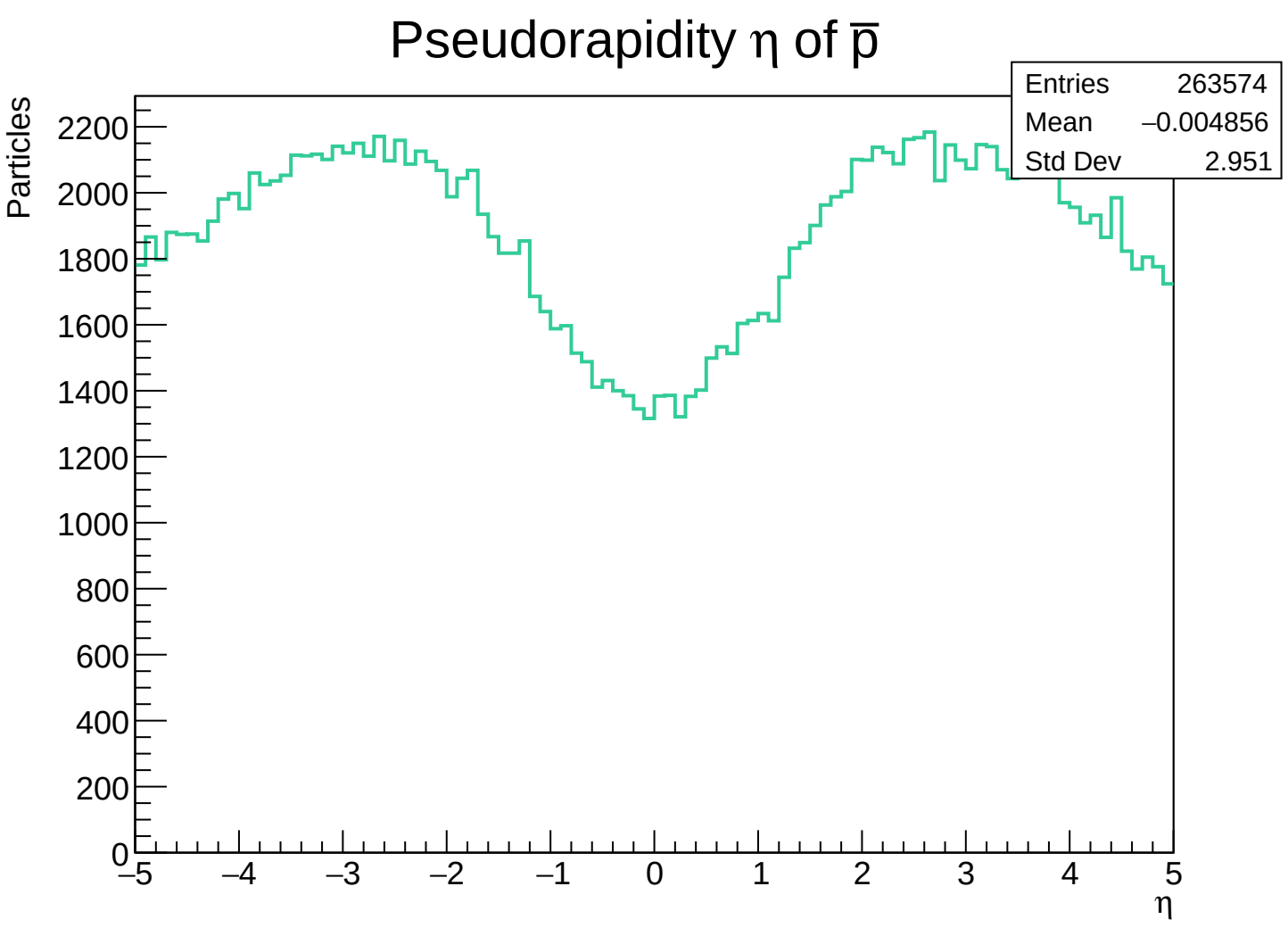

Pseudorapidity η of π+
Entries 3819114
Mean 0.000857
Std Dev 2.789
Particles
30000
25000
20000
15000
10000
5000
0
–5
–4
–3
–2
–1
0
1
2
3
4
5
η

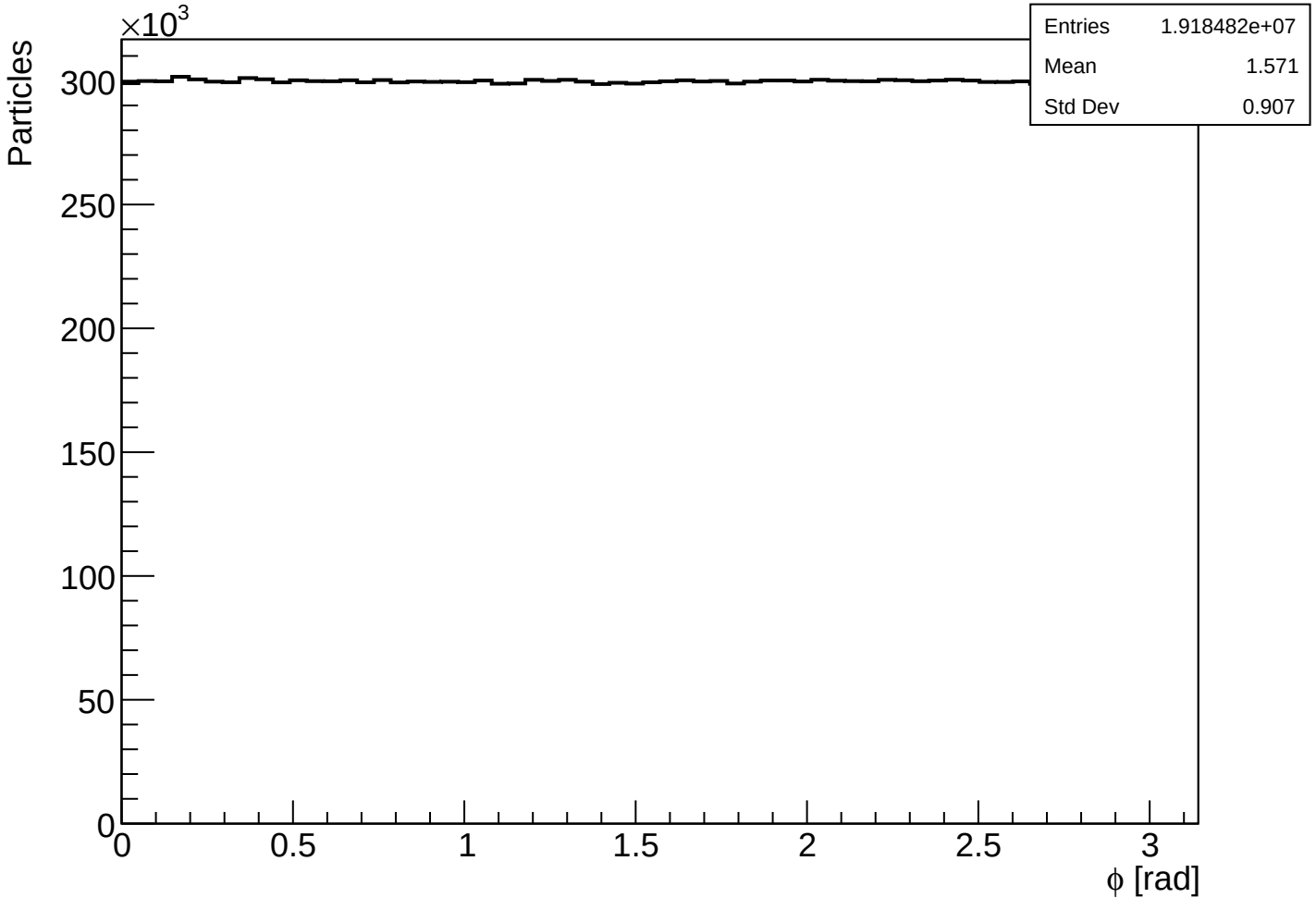

Pseudorapidity η of π-
Entries 3751339
Mean –0.0004987
Std Dev 2.788
Particles
30000
25000
20000
15000
10000
5000
0
–5
–4
–3
–2
–1
0
1
2
3
4
5
η

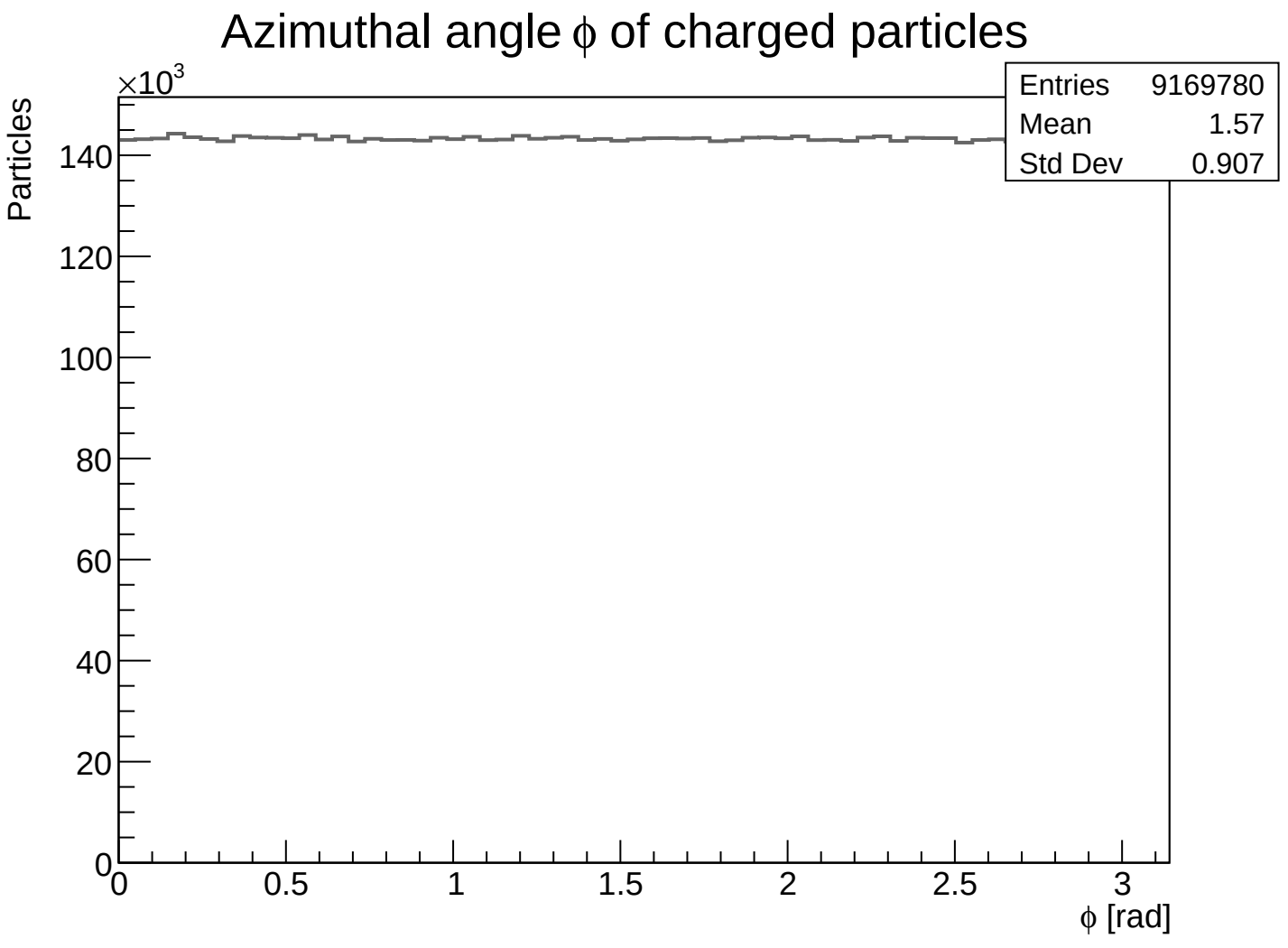

Pseudorapidity η of K+
Entries 420918
Mean –0.001085
Std Dev 2.866
Particles
3500
3000
2500
2000
1500
1000
500
0
–5
–4
–3
–2
–1
0
1
2
3
4
5
η

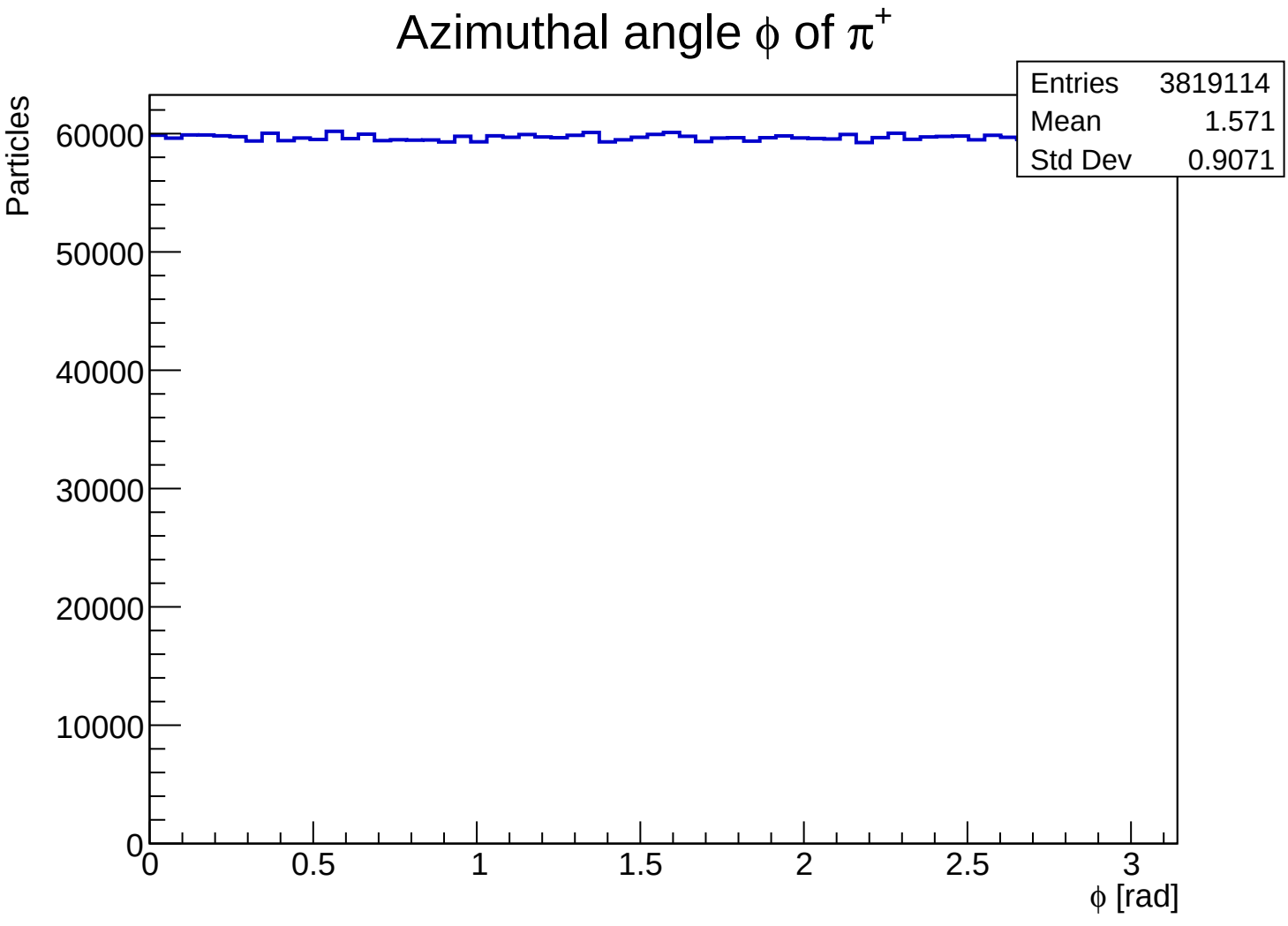

Pseudorapidity η of K-
Entries 410552
Mean 0.002391
Std Dev 2.858
Particles
3500
3000
2500
2000
1500
1000
500
0
–5
–4
–3
–2
–1
0
1
2
3
4
5
η

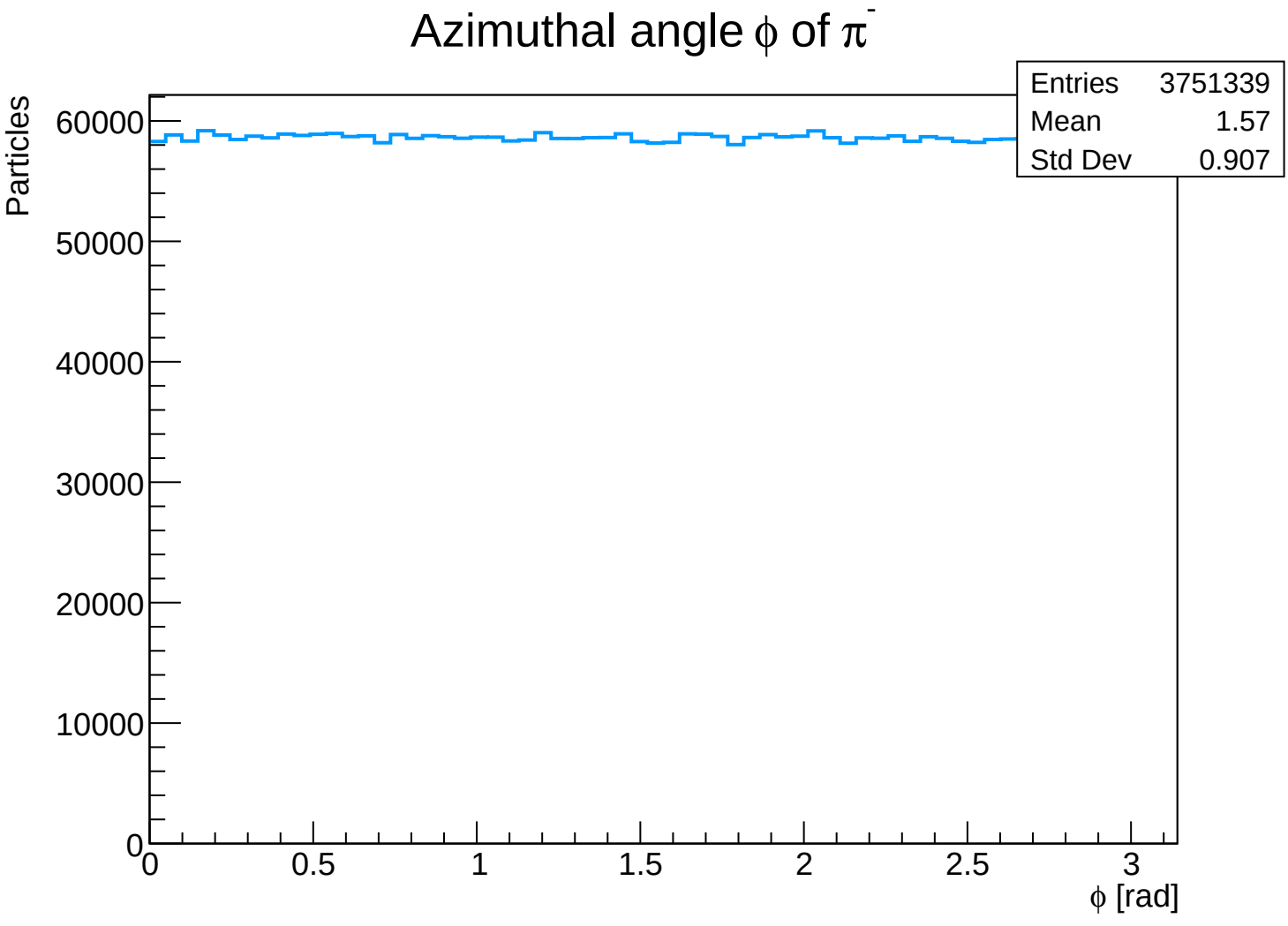

Pseudorapidity η of p
Entries 385466
Mean 0.006782
Std Dev 2.965
Particles
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
–5
–4
–3
–2
–1
0
1
2
3
4
5
η

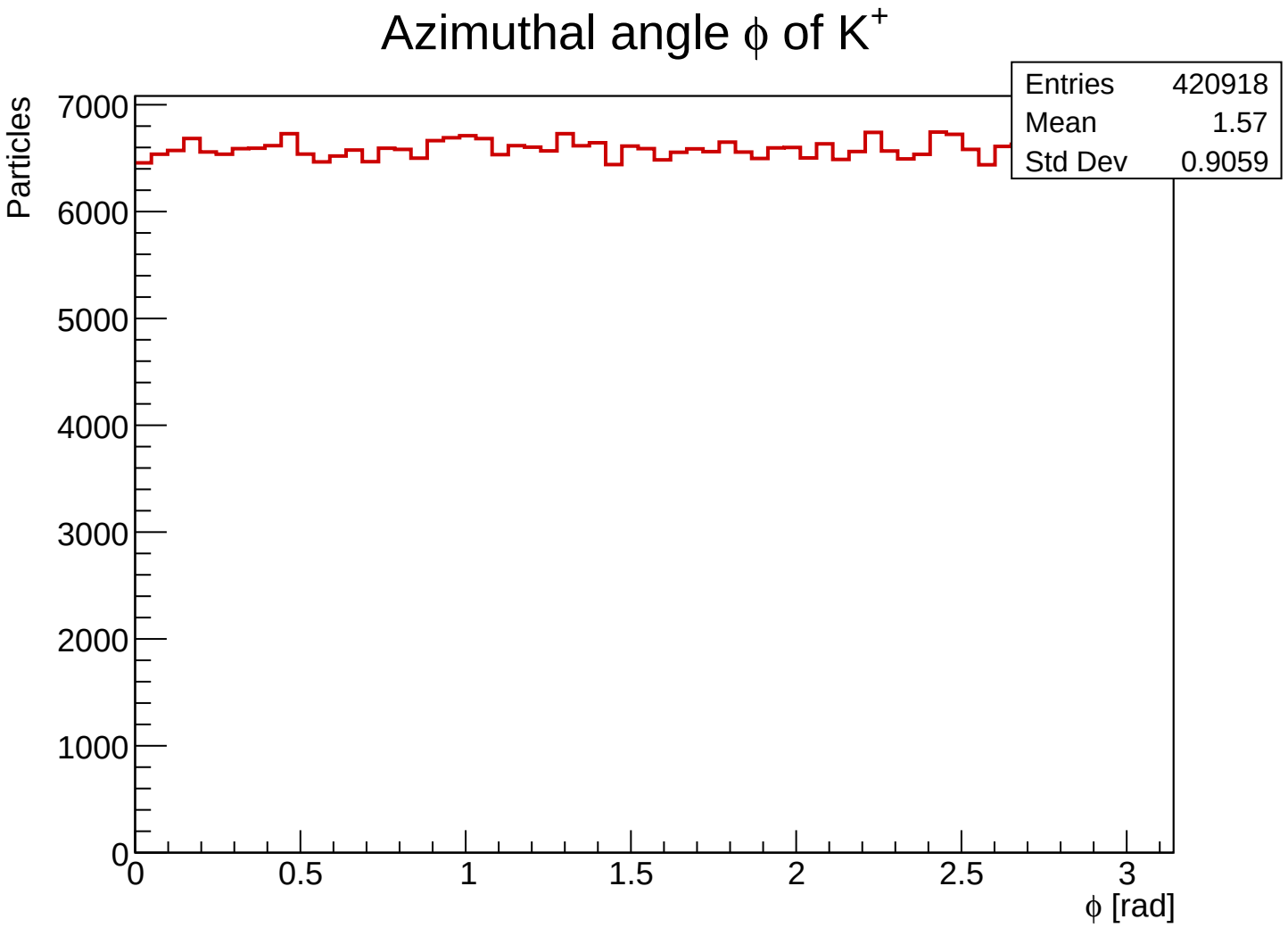

Pseudorapidity η of $\bar{p}$
Entries 263574
Mean −0.004856
Std Dev 2.951
Particles
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−5
−4
−3
−2
−1
0
1
2
3
4
5
η

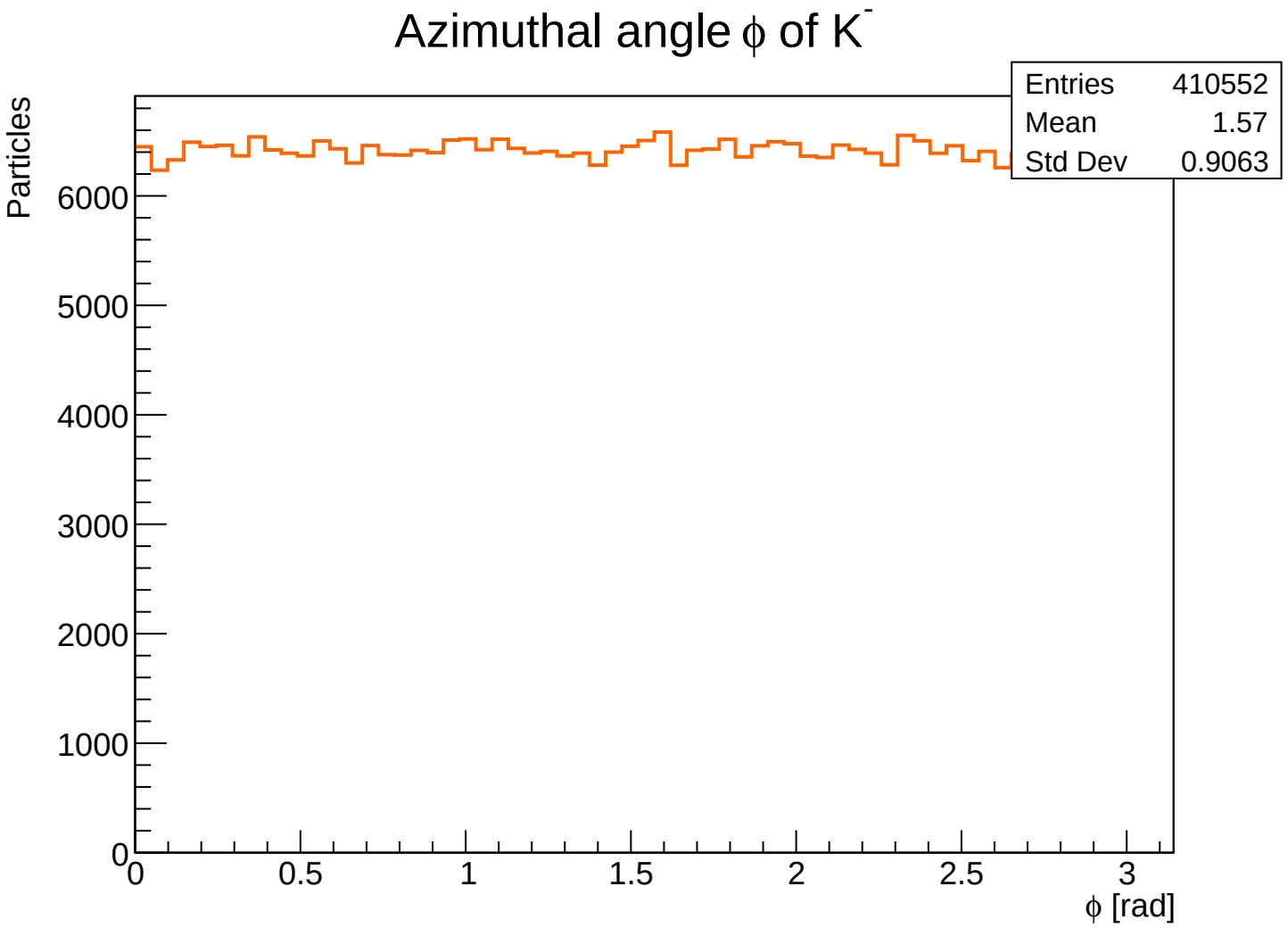

Azimuthal angle ϕ of all final-state particles
Entries 1.918482e+07
Mean 1.571
Std Dev 0.907
×10^3
Particles
300
250
200
150
100
50
0
0
0.5
1
1.5
2
2.5
3
ϕ [rad]

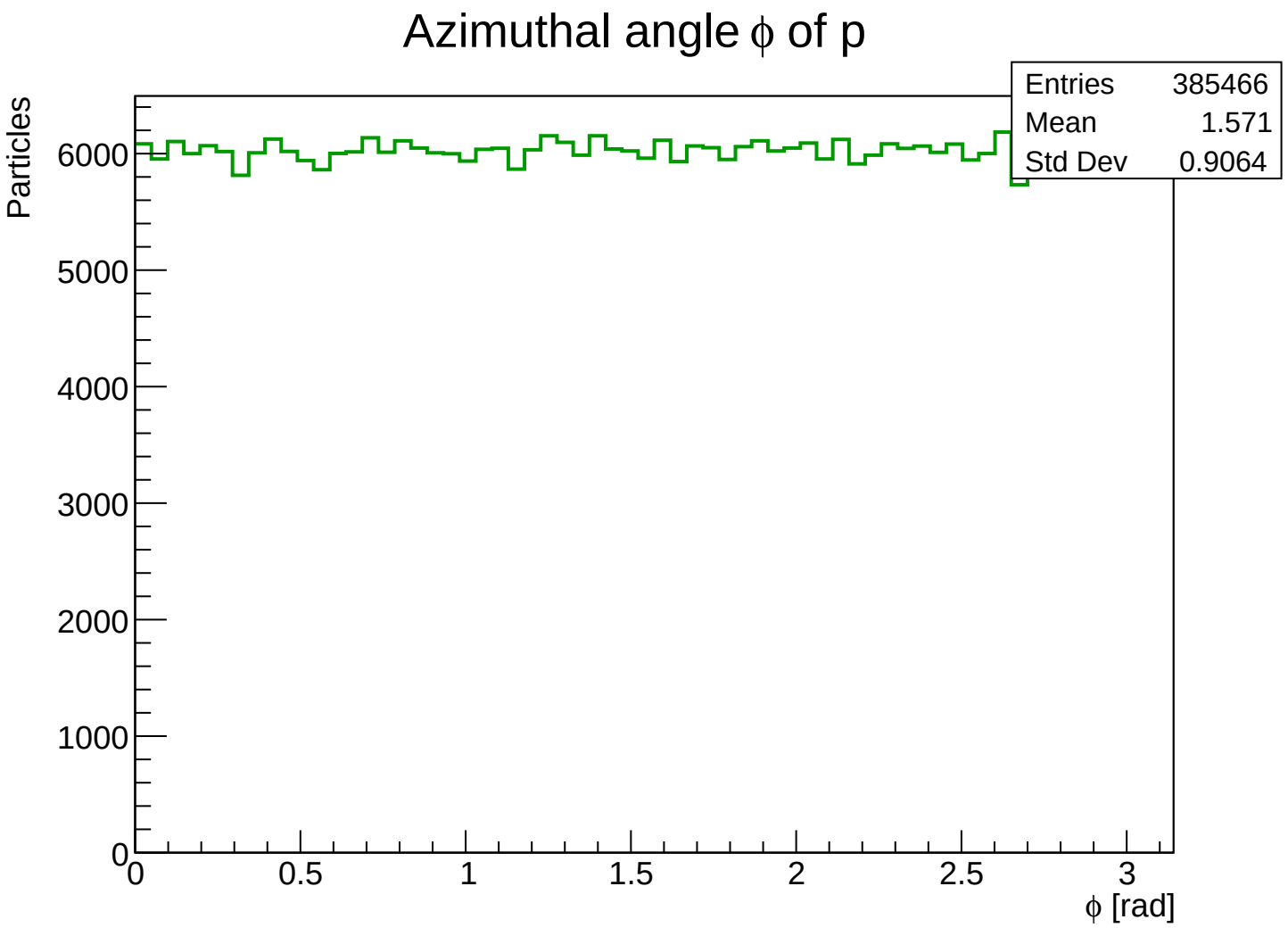

Azimuthal angle ϕ of charged particles
Entries 9169780
Mean 1.57
Std Dev 0.907
×10^3
Particles
140
120
100
80
60
40
20
0
0
0.5
1
1.5
2
2.5
3
ϕ [rad]

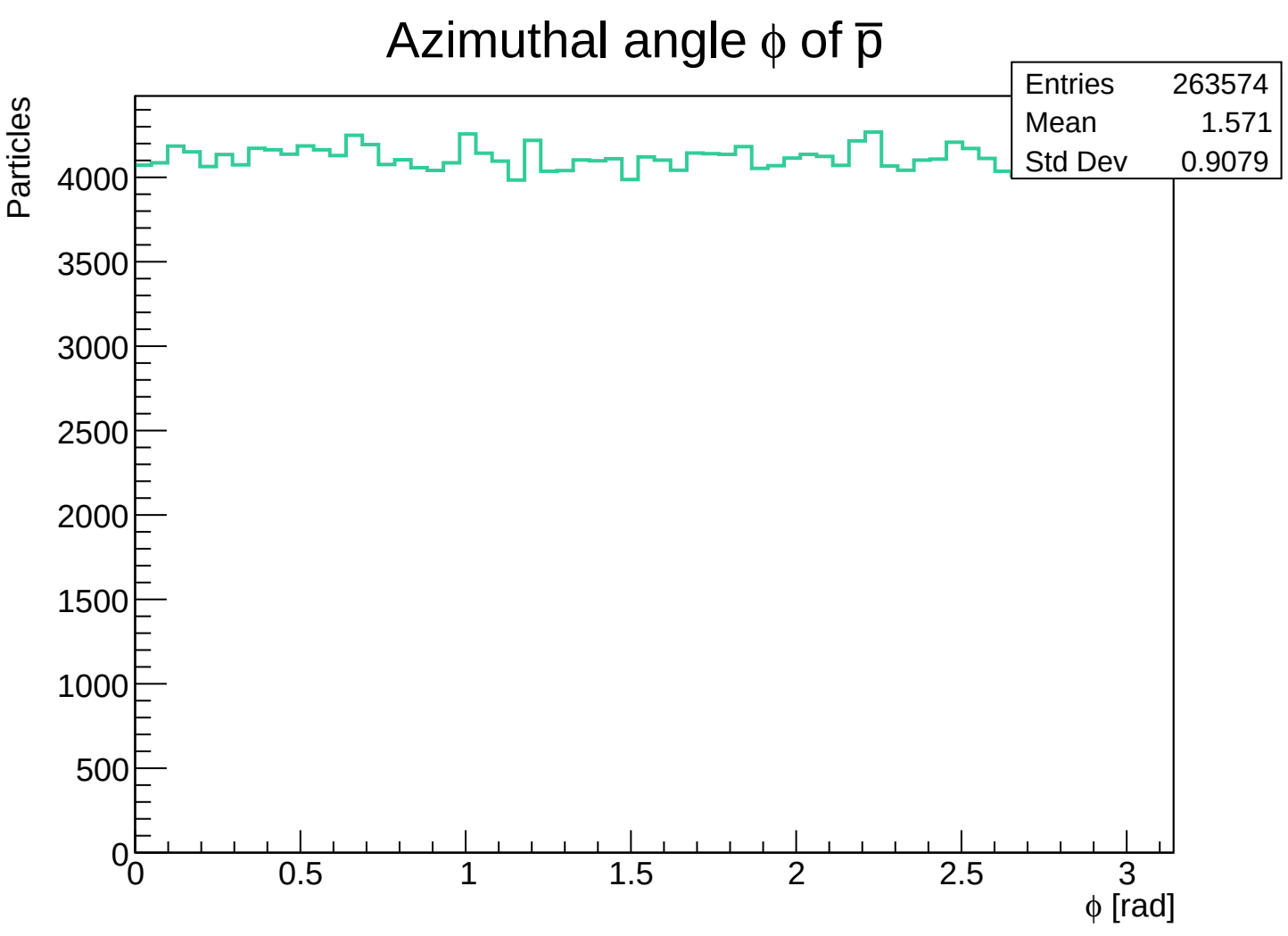

Azimuthal angle ϕ of π+
Entries 3819114
Mean 1.571
Std Dev 0.9071
Particles
60000
50000
40000
30000
20000
10000
0
0
0.5
1
1.5
2
2.5
3
ϕ [rad]

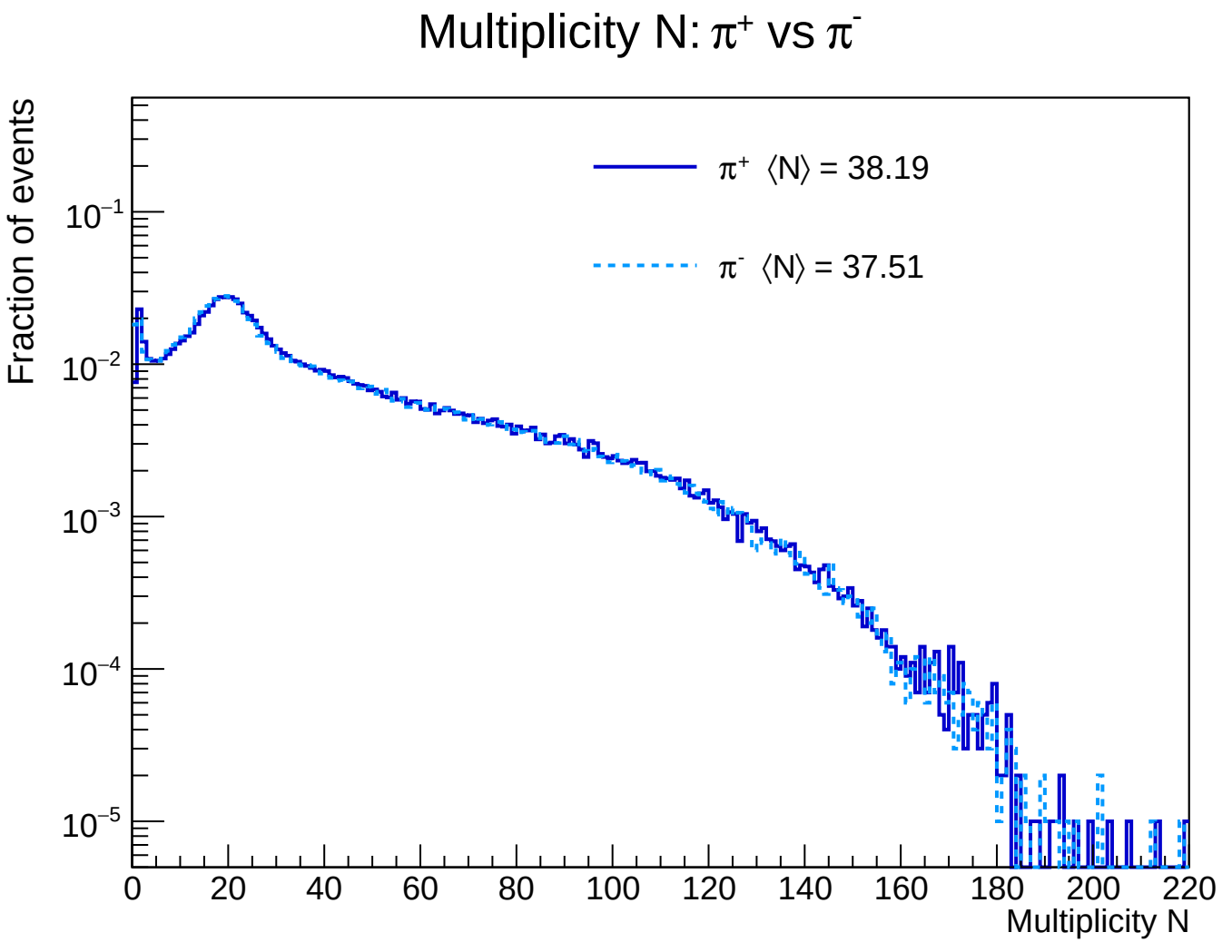

Azimuthal angle ϕ of π-
Entries 3751339
Mean 1.57
Std Dev 0.907
Particles
60000
50000
40000
30000
20000
10000
0
0
0.5
1
1.5
2
2.5
3
ϕ [rad]

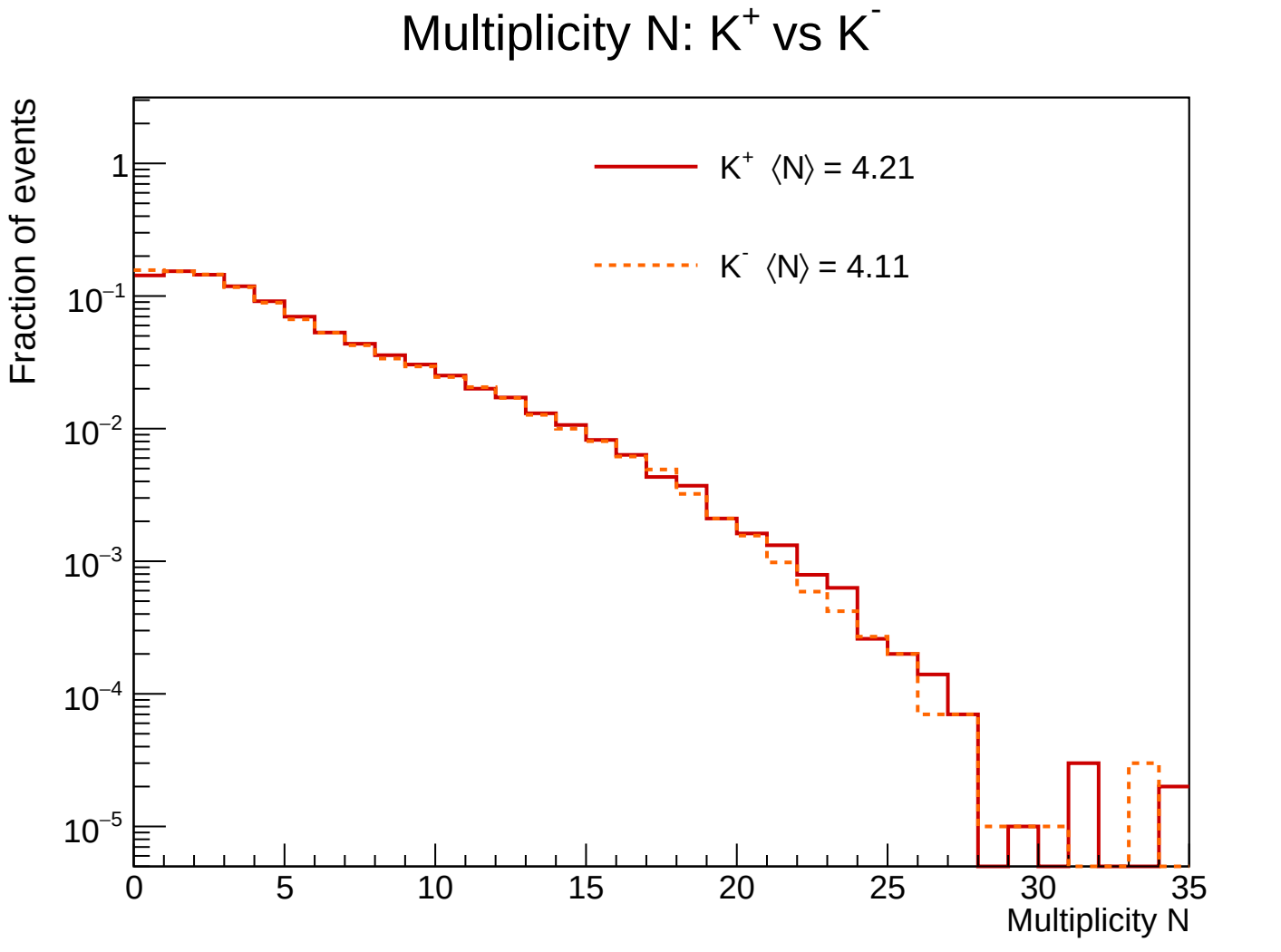

Azimuthal angle ϕ of K+
Entries 420918
Mean 1.57
Std Dev 0.9059
Particles
7000
6000
5000
4000
3000
2000
1000
0
0
0.5
1
1.5
2
2.5
3
ϕ [rad]

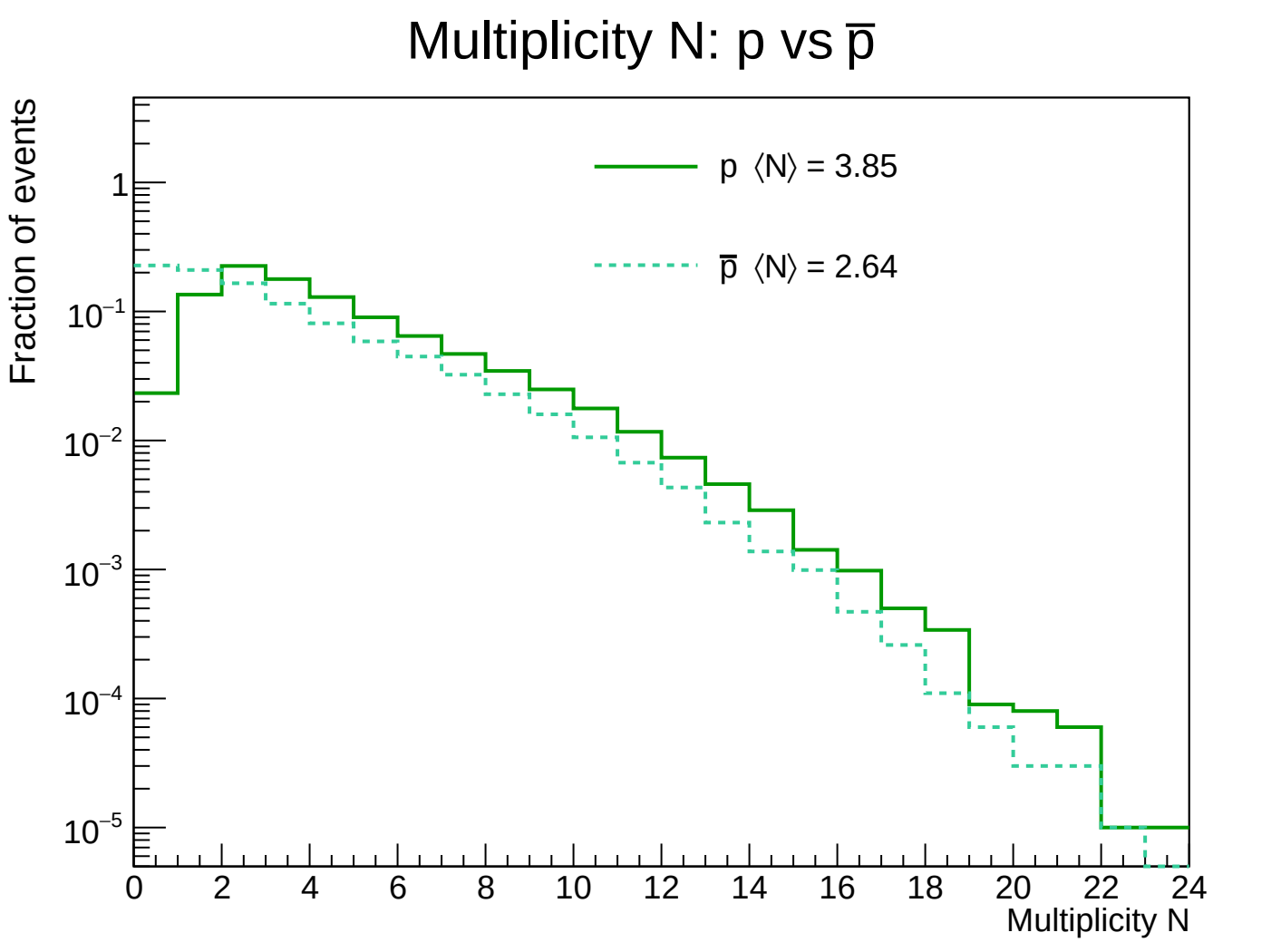

Azimuthal angle ϕ of K-
Entries 410552
Mean 1.57
Std Dev 0.9063
Particles
6000
5000
4000
3000
2000
1000
0
0
0.5
1
1.5
2
2.5
3
ϕ [rad]

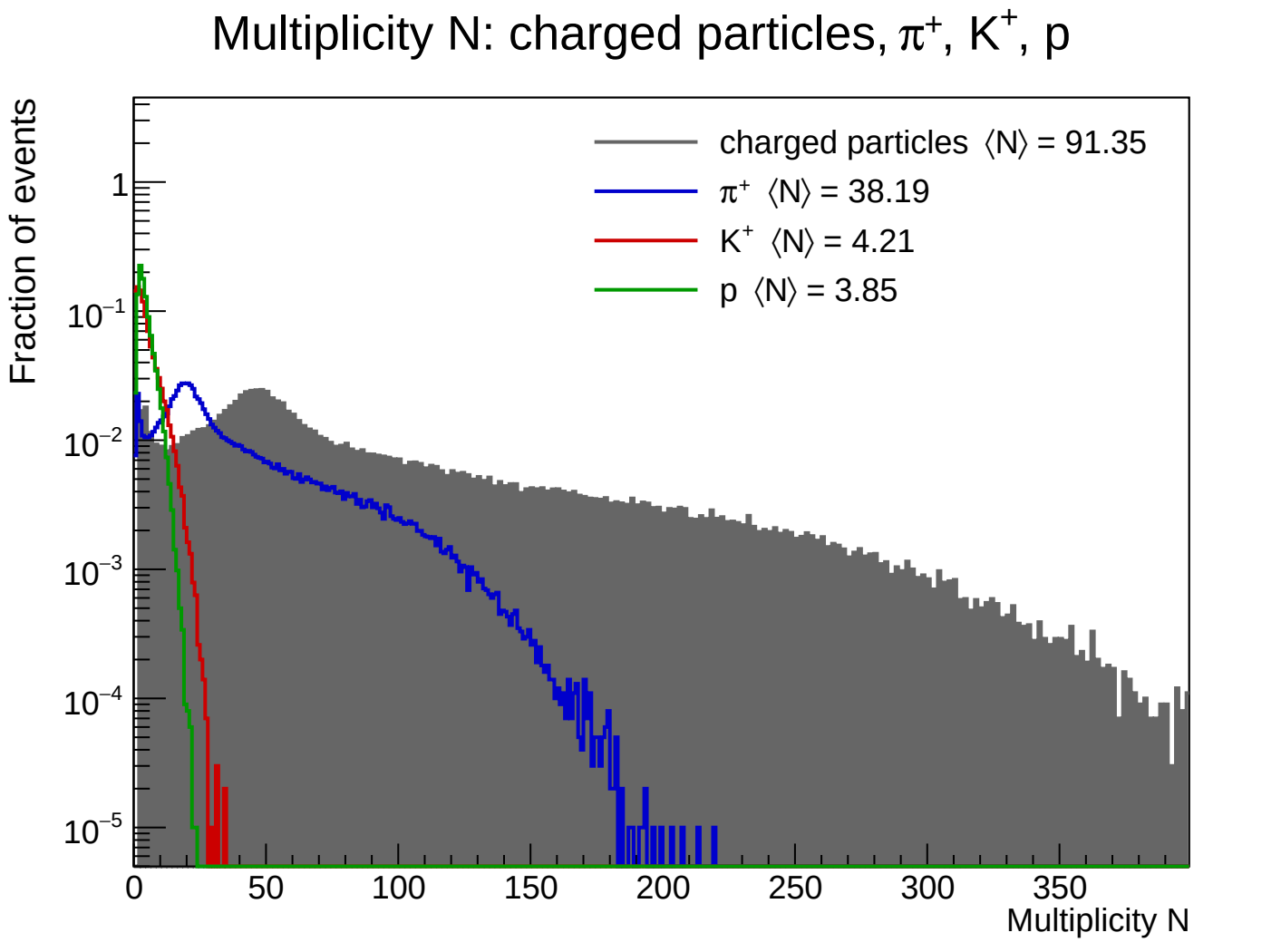

Azimuthal angle ϕ of p
Entries 385466
Mean 1.571
Std Dev 0.9064
Particles
6000
5000
4000
3000
2000
1000
0
0
0.5
1
1.5
2
2.5
3
ϕ [rad]

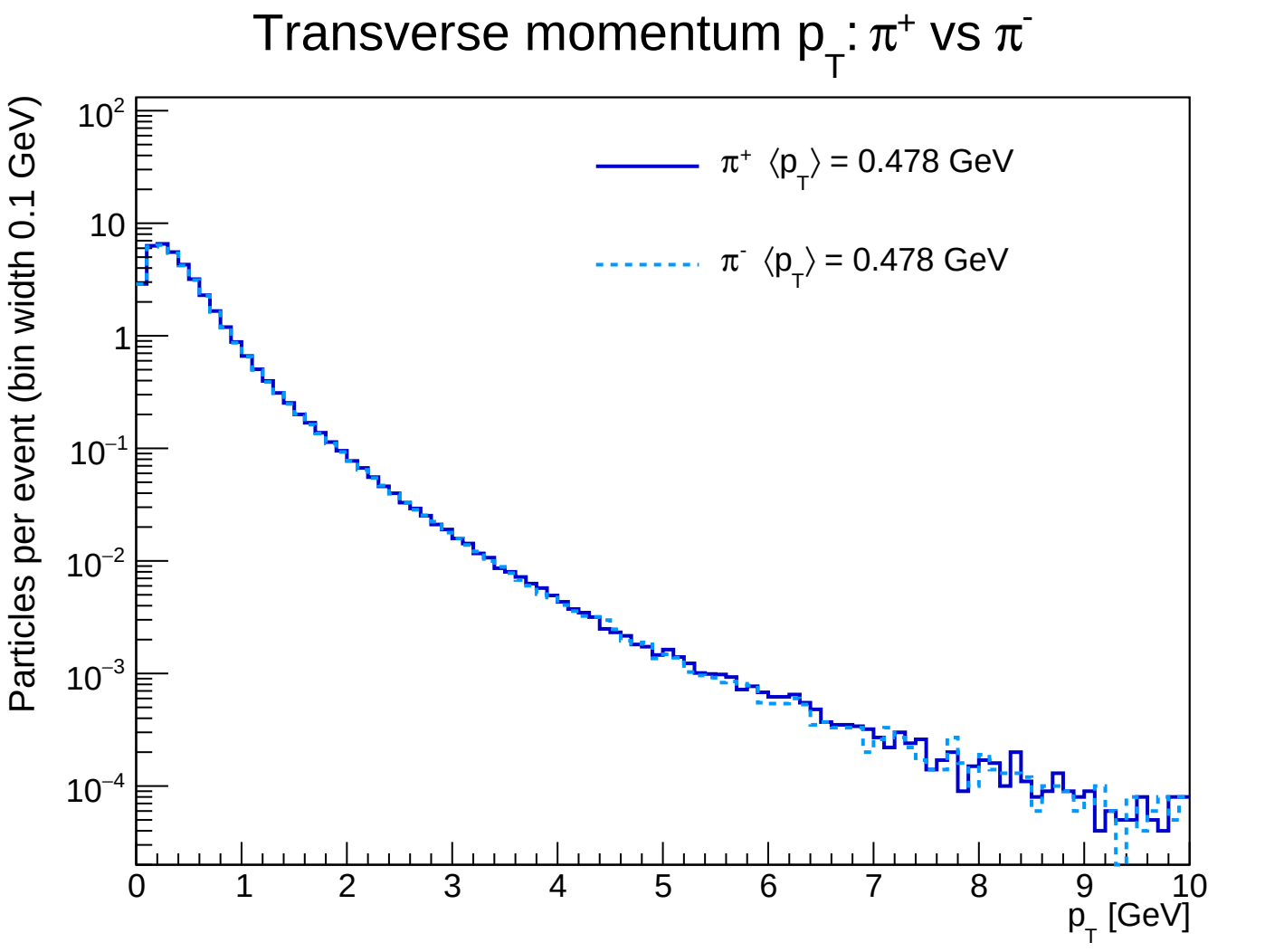

Azimuthal angle ϕ of p̄
Entries 263574
Mean 1.571
Std Dev 0.9079
Particles
4000
3500
3000
2500
2000
1500
1000
500
0
0
0.5
1
1.5
2
2.5
3
ϕ [rad]

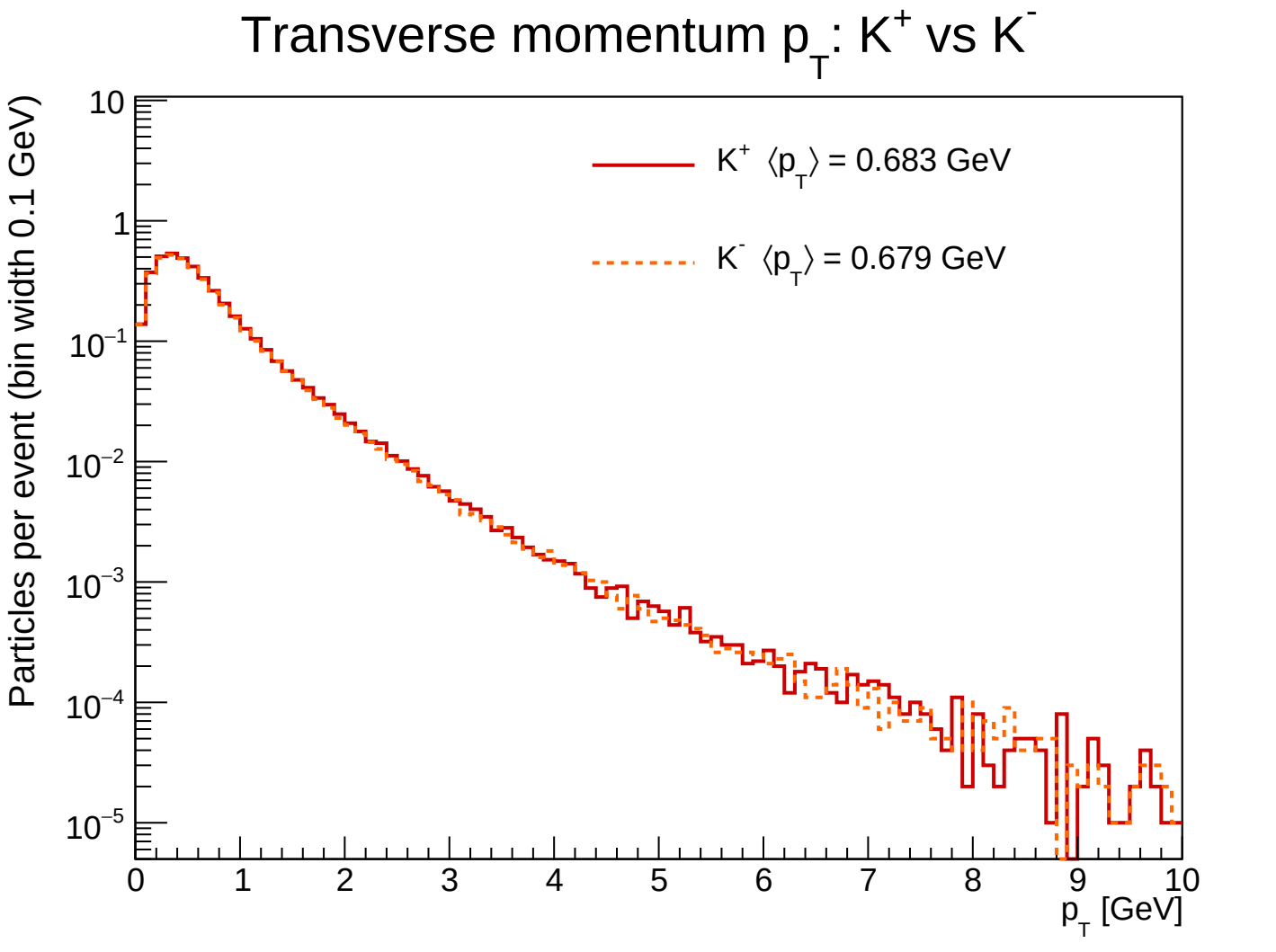

Multiplicity N: π+ vs π-
π+ ⟨N⟩ = 38.19
π- ⟨N⟩ = 37.51
Fraction of events
10−1
10−2
10−3
10−4
10−5
0
20
40
60
80
100
120
140
160
180
200
220
Multiplicity N

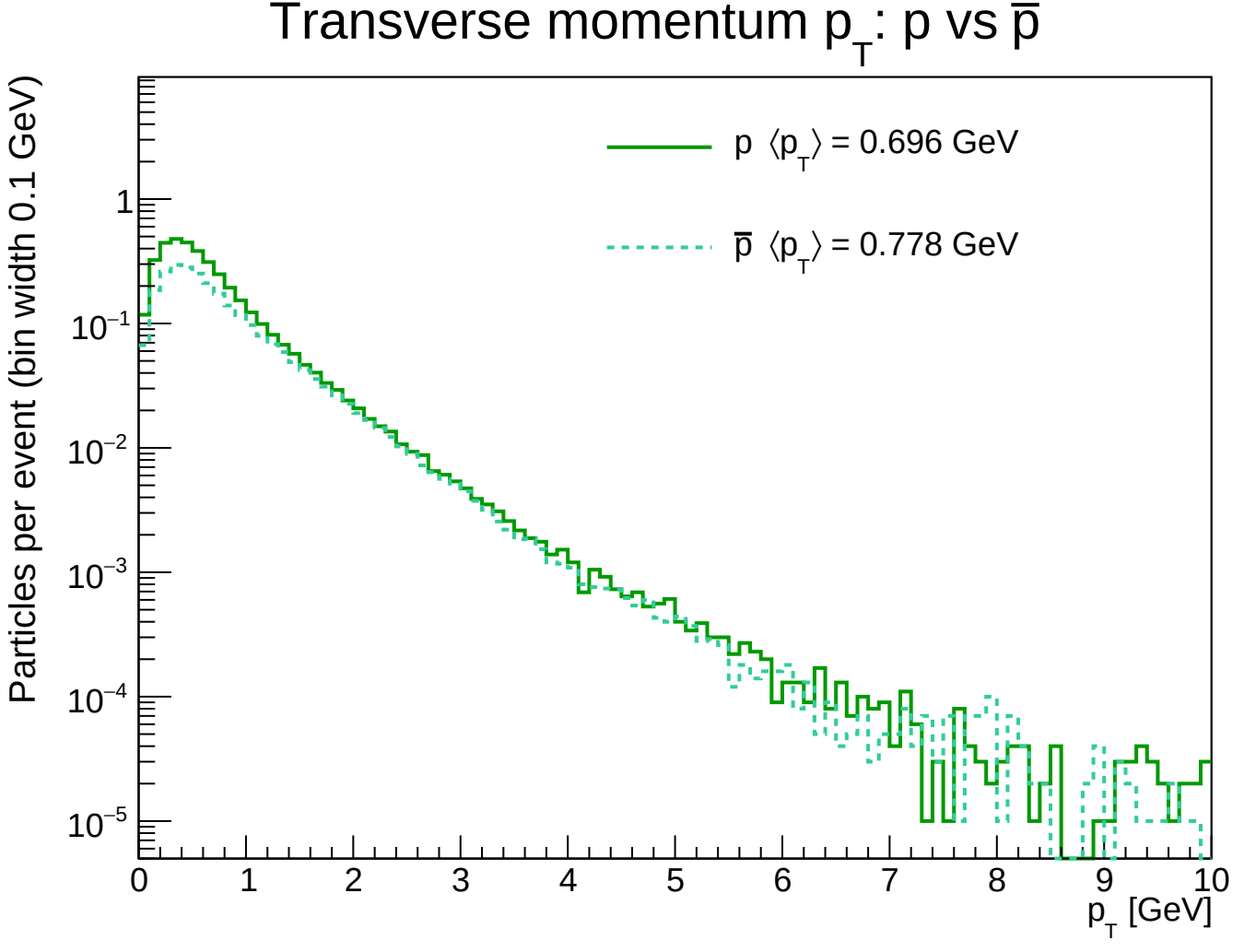

Multiplicity N: K+ vs K-
Fraction of events
1
10−1
10−2
10−3
10−4
10−5
K+ ⟨N⟩ = 4.21
K- ⟨N⟩ = 4.11
0
5
10
15
20
25
30
35
Multiplicity N

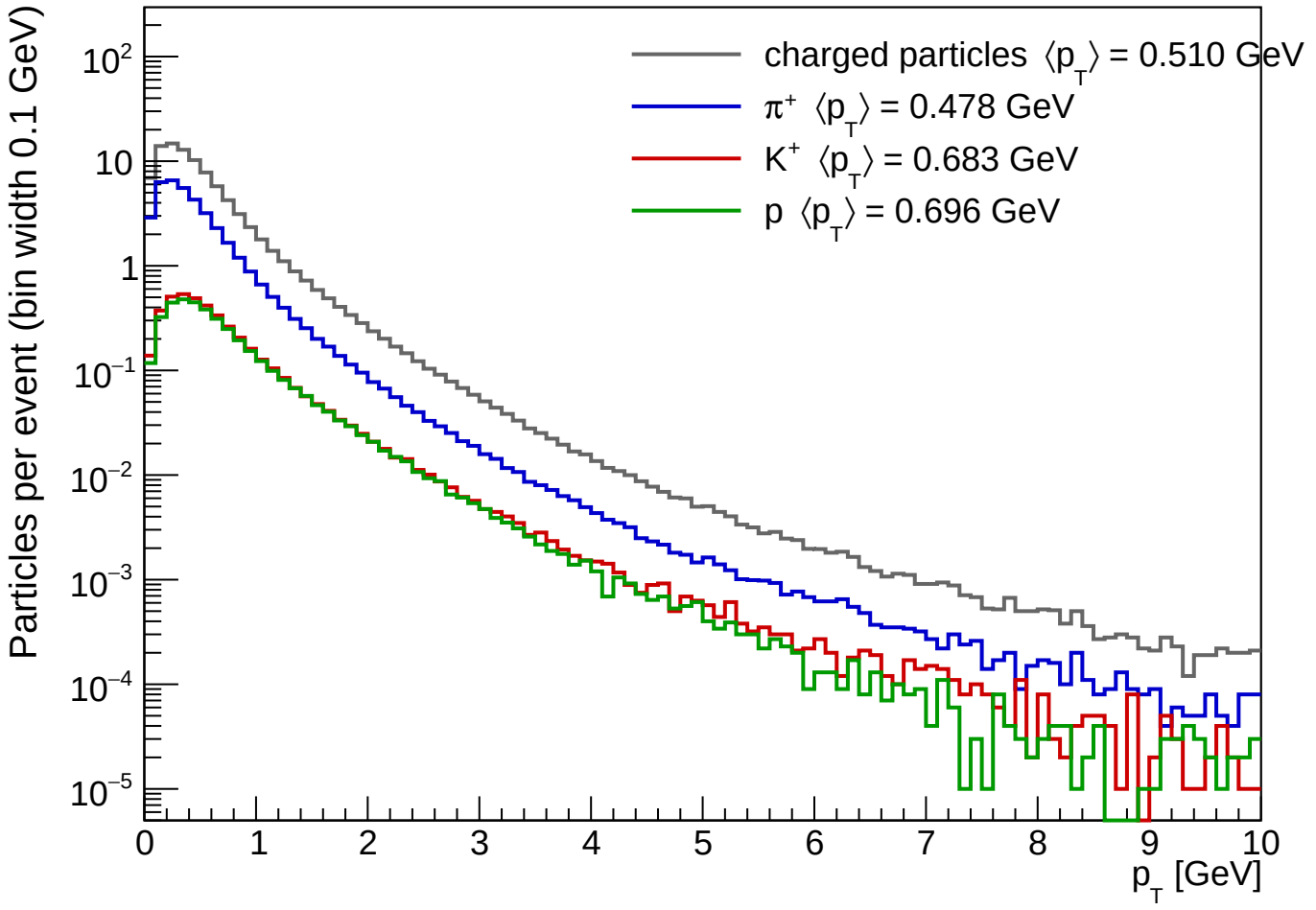

Multiplicity N: p vs $\bar{p}$
Fraction of events
1
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
p ⟨N⟩ = 3.85
$\bar{p}$ ⟨N⟩ = 2.64
0
2
4
6
8
10
12
14
16
18
20
22
24
Multiplicity N

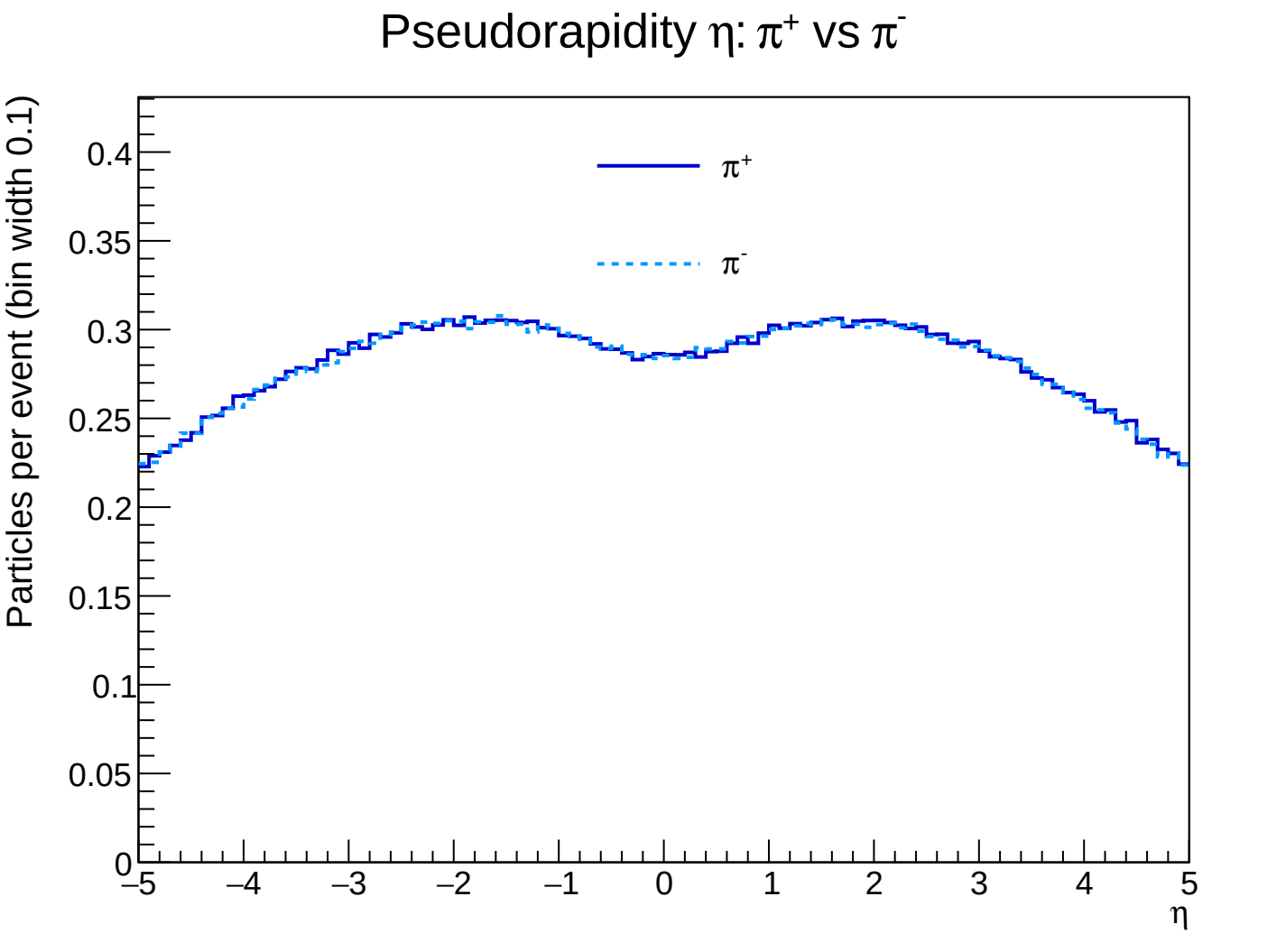

Multiplicity N: charged particles, π+, K+, p
charged particles ⟨N⟩ = 91.35
π+ ⟨N⟩ = 38.19
K+ ⟨N⟩ = 4.21
p ⟨N⟩ = 3.85
Fraction of events
1
10−1
10−2
10−3
10−4
10−5
0
50
100
150
200
250
300
350
Multiplicity N

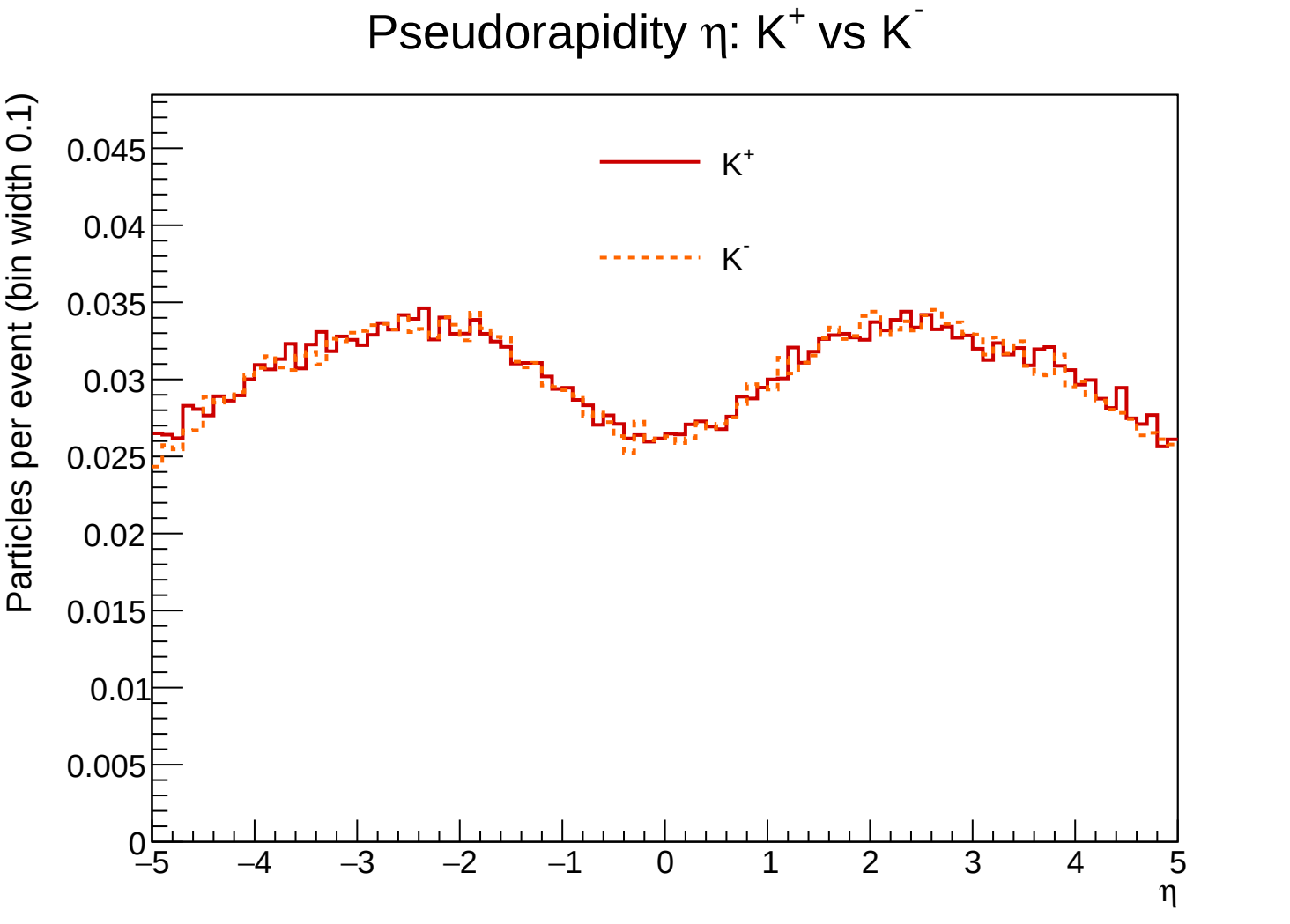

Transverse momentum p_T: π^+ vs π^-
π^+ $\langle p_T \rangle$ = 0.478 GeV
π^- $\langle p_T \rangle$ = 0.478 GeV
Particles per event (bin width 0.1 GeV)
p_T [GeV]
10^2
10
1
10^{-1}
10^{-2}
10^{-3}
10^{-4}
0
1
2
3
4
5
6
7
8
9
10

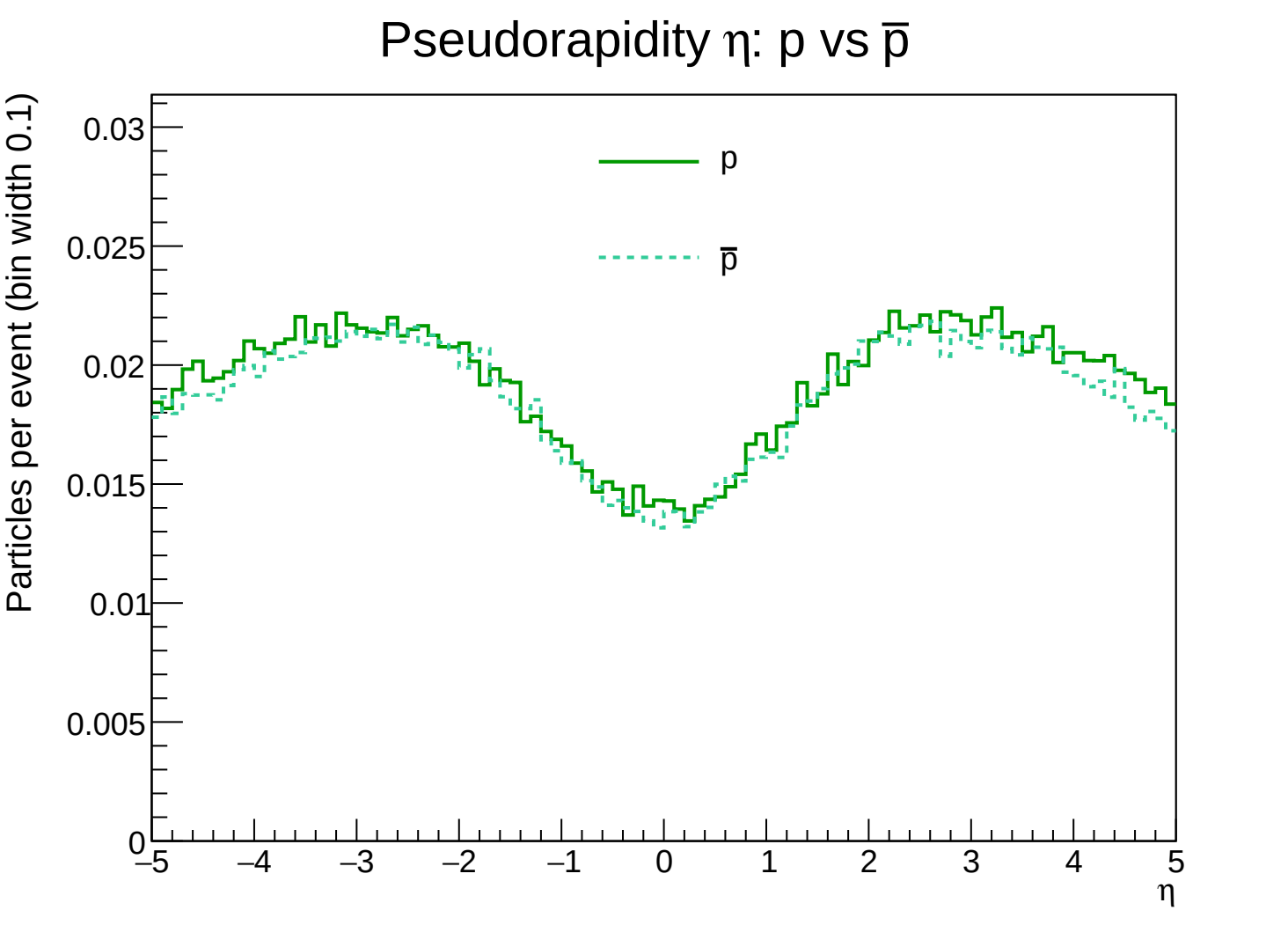

Transverse momentum p_T: K^+ vs K^-
K^+ $\langle p_T \rangle$ = 0.683 GeV
K^- $\langle p_T \rangle$ = 0.679 GeV
Particles per event (bin width 0.1 GeV)
p_T [GeV]
10
1
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
1
2
3
4
5
6
7
8
9
10

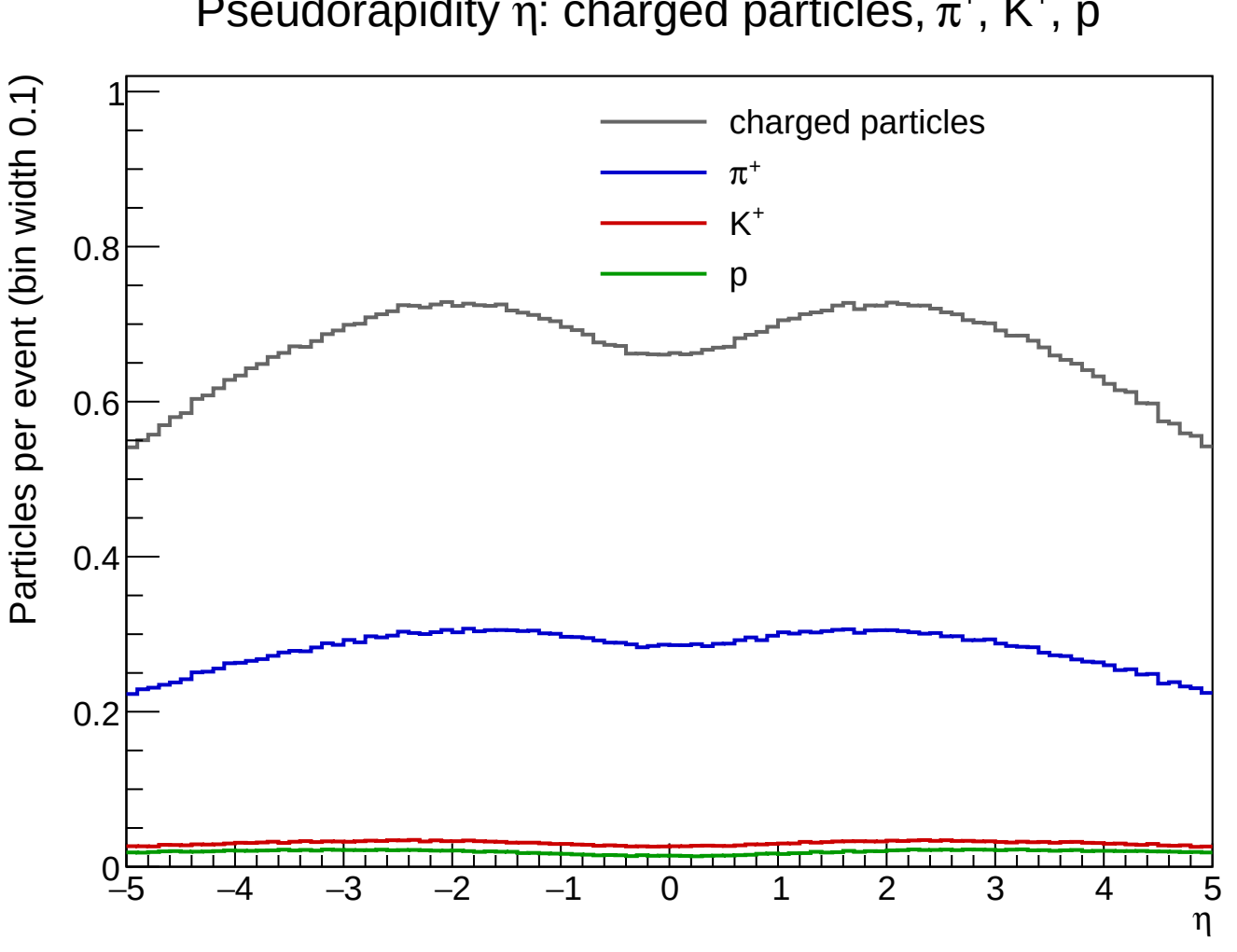

Transverse momentum p_T: p vs $\bar{p}$
p $\langle p_T \rangle$ = 0.696 GeV
$\bar{p}$ $\langle p_T \rangle$ = 0.778 GeV
Particles per event (bin width 0.1 GeV)
1
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
1
2
3
4
5
6
7
8
9
10
p_T [GeV]

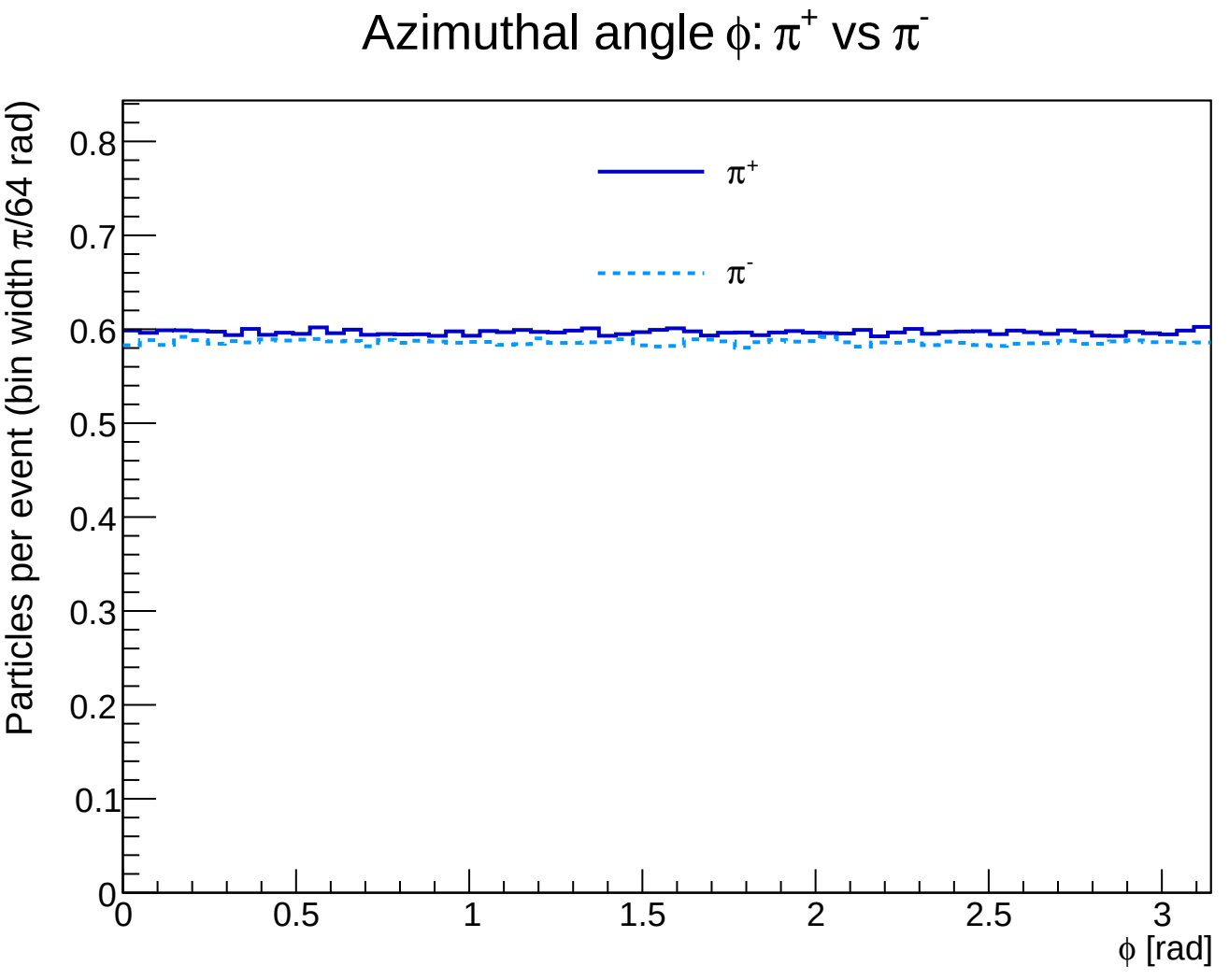

Transverse momentum p_T: charged particles, π^+, K^+, p
charged particles $\langle p_T \rangle$ = 0.510 GeV
π^+ $\langle p_T \rangle$ = 0.478 GeV
K^+ $\langle p_T \rangle$ = 0.683 GeV
p $\langle p_T \rangle$ = 0.696 GeV
Particles per event (bin width 0.1 GeV)
10^2
10
1
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
1
2
3
4
5
6
7
8
9
10
p_T [GeV]

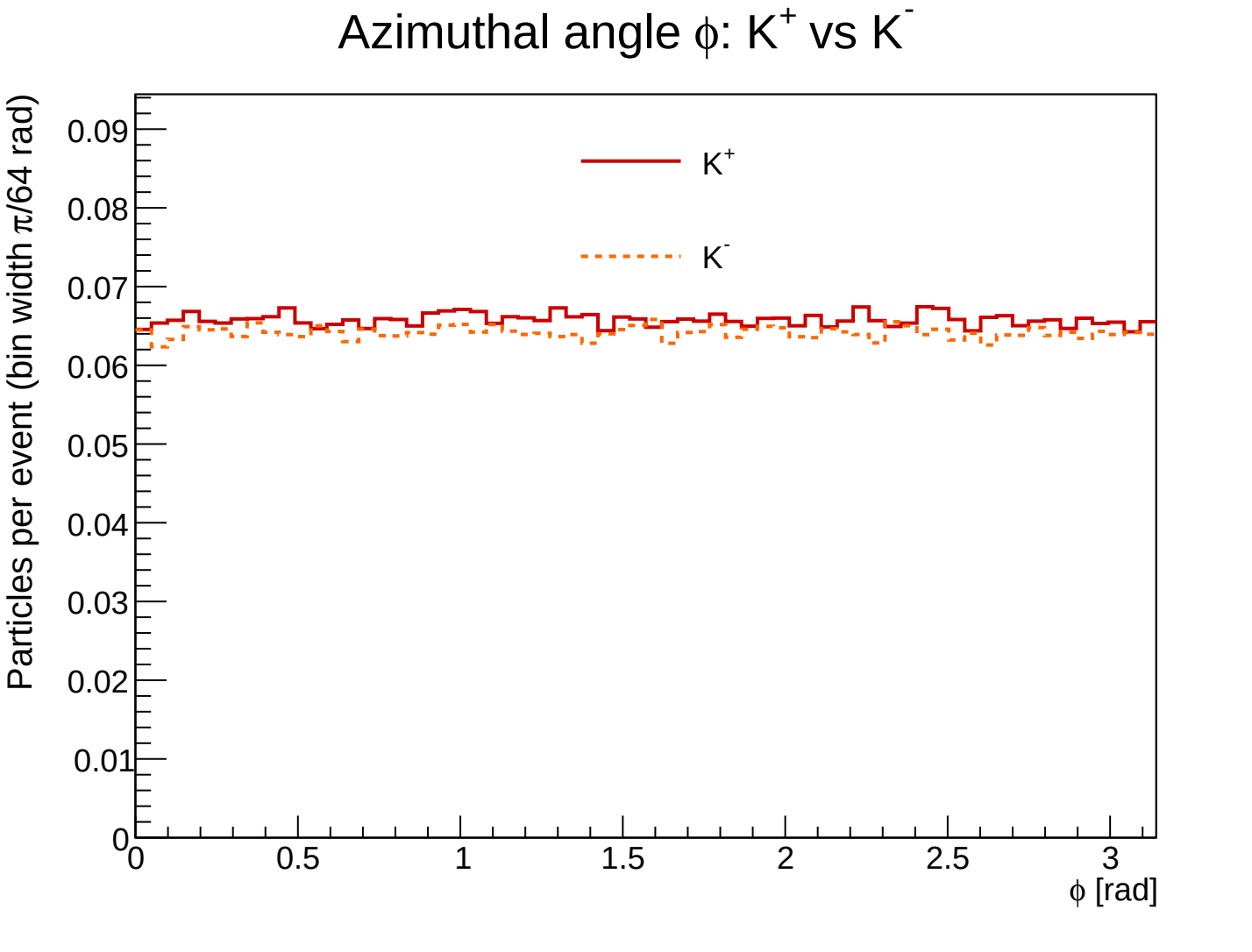

Pseudorapidity η: π+ vs π-
Particles per event (bin width 0.1)
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
π+
π-
–5
–4
–3
–2
–1
0
1
2
3
4
5
η

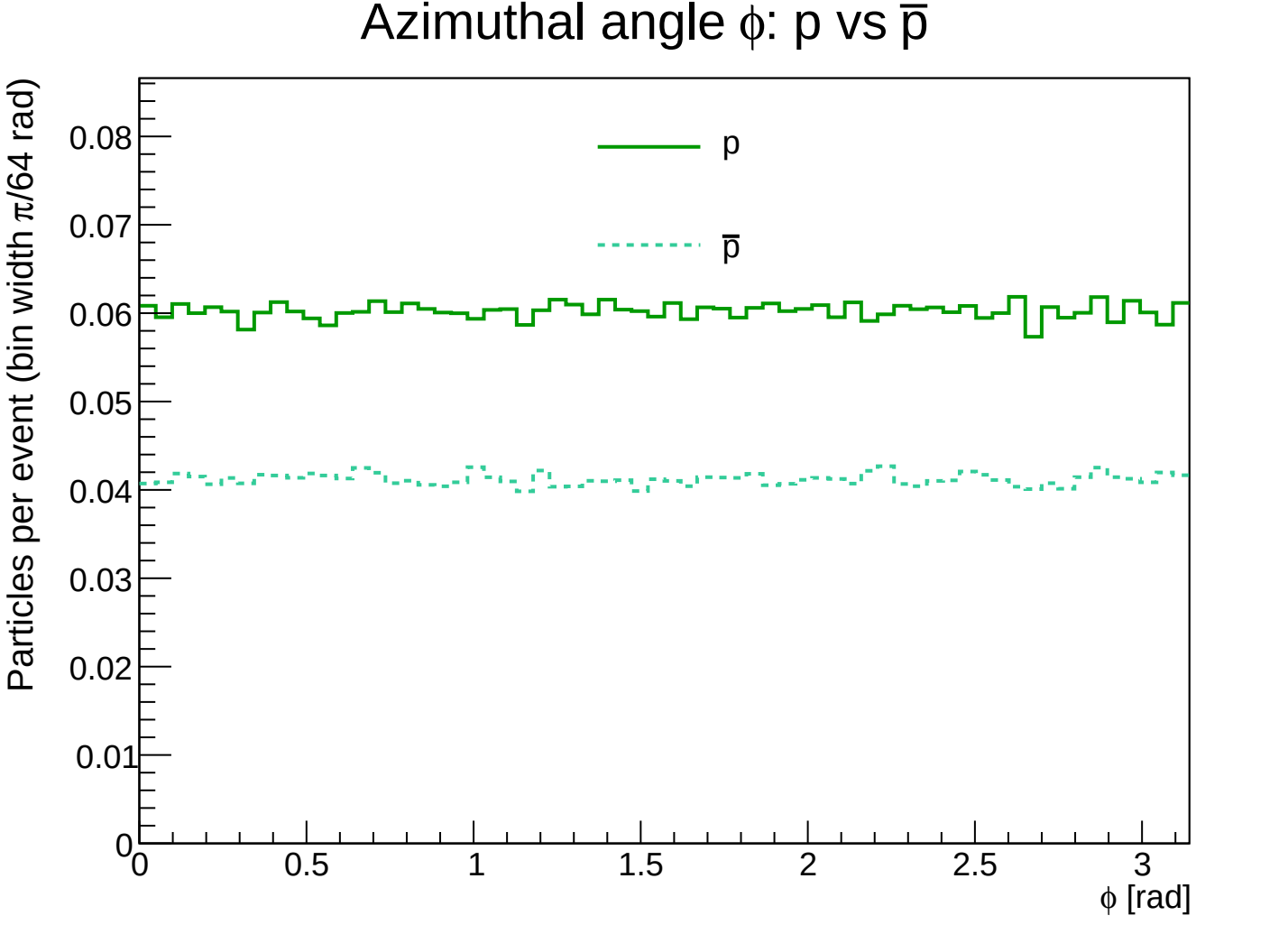

Pseudorapidity η: K+ vs K-
Particles per event (bin width 0.1)
0.045
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
K+
K-
–5
–4
–3
–2
–1
0
1
2
3
4
5
η

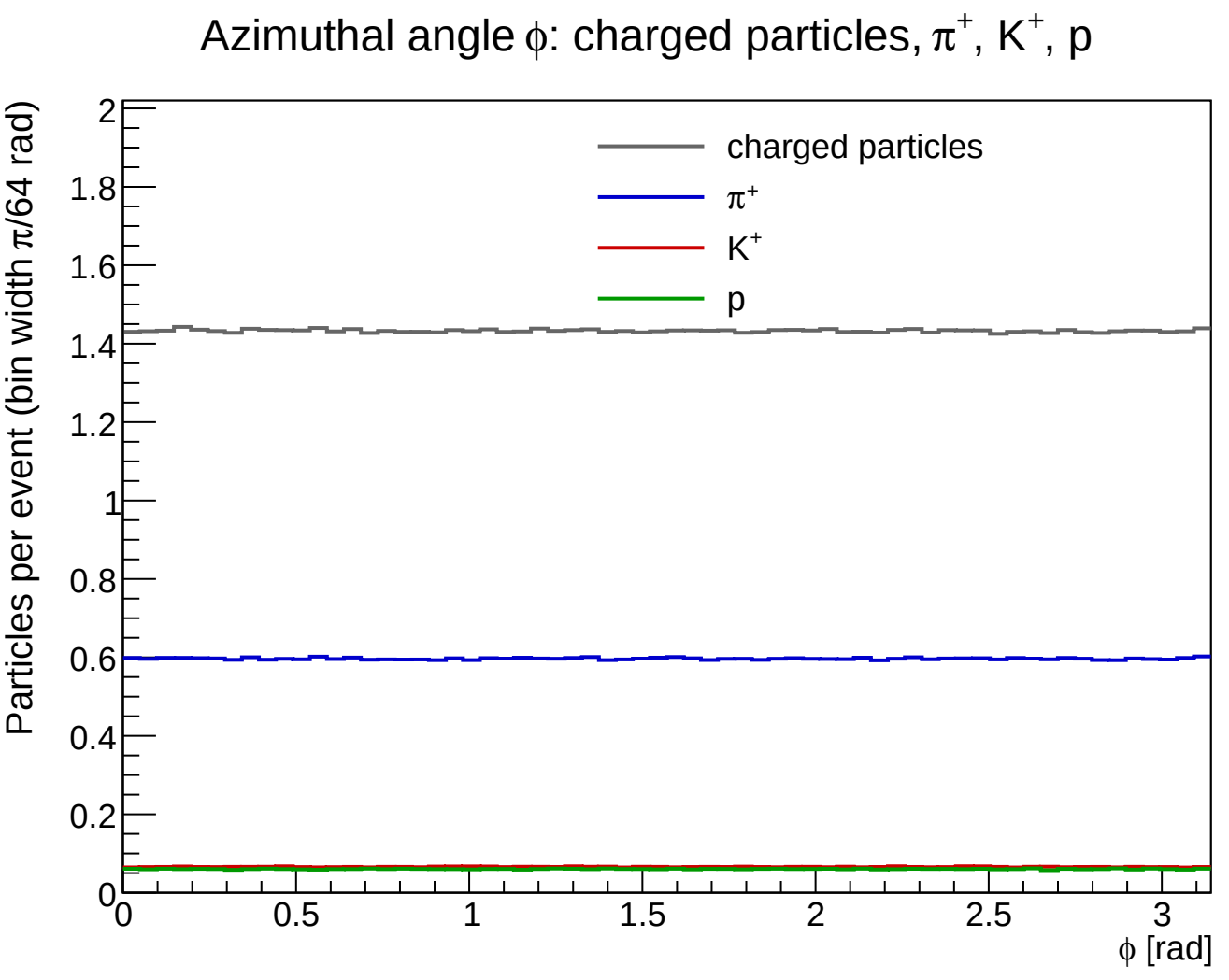

Pseudorapidity η: p vs p̄
Particles per event (bin width 0.1)
0.03
0.025
0.02
0.015
0.01
0.005
0
p
p̄
–5
–4
–3
–2
–1
0
1
2
3
4
5
η


Pseudorapidity η: charged particles, π+, K+, p
Particles per event (bin width 0.1)
charged particles
π+
K+
p
1
0.8
0.6
0.4
0.2
0
–5
–4
–3
–2
–1
0
1
2
3
4
5
η


Azimuthal angle ϕ: π+ vs π-
Particles per event (bin width π/64 rad)
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0
0.5
1
1.5
2
2.5
3
ϕ [rad]
π+
π-


Azimuthal angle ϕ: K+ vs K-
Particles per event (bin width π/64 rad)
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
0
0.5
1
1.5
2
2.5
3
ϕ [rad]
K+
K-


Azimuthal angle ϕ: p vs p̄
Particles per event (bin width π/64 rad)
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
p
p̄
0
0.5
1
1.5
2
2.5
3
ϕ [rad]


Azimuthal angle ϕ: charged particles, π+, K+, p
charged particles
π+
K+
p
Particles per event (bin width π/64 rad)
ϕ [rad]
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0
0.5
1
1.5
2
2.5
3